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MINIMAX RATES FOR CONDITIONAL DENSITY ESTIMATION VIA EMPIRICAL ENTROPY

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We consider the task of estimating a conditional density using i.i.d. samples from a joint distribution, which is a fundamental problem with applications in both classification and uncertainty quantification for regression. For joint density estimation, minimax rates have been characterized for general density classes in terms of uniform (metric) entropy, a well-studied notion of statistical capacity. When applying these results to conditional density estimation, the use of uniform entropy—which is infinite when the covariate space is unbounded and suffers from the curse of dimensionality—can lead to suboptimal rates. Consequently, minimax rates for conditional density estimation cannot be characterized using these classical results.

We resolve this problem for well-specified models, obtaining matching (within logarithmic factors) upper and lower bounds on the minimax Kullback–Leibler risk in terms of the *empirical Hellinger entropy* for the conditional density class. The use of empirical entropy allows us to appeal to concentration arguments based on local Rademacher complexity, which—in contrast to uniform entropy—leads to matching rates for large, potentially nonparametric classes and captures the correct dependence on the complexity of the covariate space. Our results require only that the conditional densities are bounded above, and do not require that they are bounded below or otherwise satisfy any tail conditions.

1. Introduction. Estimating the density underlying an observed stochastic process is a central problem in statistics. We consider the task of *conditional density estimation*, a fundamental and well-studied variant of the density estimation problem in which the observed stochastic process has a natural decomposition into *covariates* (representing contextual information) and *responses*; special cases include real-valued responses (e.g., regression tasks) and binary or categorical responses (e.g., classification tasks) [6, 65, 75, 79]. Compared to nonparametric regression, where the goal is to estimate the conditional mean, estimating the conditional density allows the user to quantify uncertainty, which can be used to reliably inform downstream decisions. For example, applications to online decision making may be found in Agarwal et al. [1] and Foster and Krishnamurthy [34]. However, while the task of *joint* density estimation (i.e., estimating the density for both covariates and responses jointly) has enjoyed extensive classical investigation, conditional density estimation presents unique technical challenges, and optimal estimators and rates of convergence for general conditional density classes are not well understood.

We focus on conditional density estimation under *logarithmic loss*, and aim to minimize the associated *Kullback–Leibler* (KL) risk

$$\mathbb{E}_{x\sim\mu_{\mathcal{X}}}\,\mathrm{KL}\big(f^*(x)\,\|\,\hat{f}(x)\big),$$

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where $\mu_{\mathcal{X}}$ is the (unknown) marginal distribution over covariates, f^* is the (unknown) true conditional density function, and $\hat{f}$ is the estimator. Classical results in (joint) density estimation and nonparametric regression [11, 12, 54, 80] have established that under fairly general conditions, the minimax rates of convergence for these tasks are determined (up to logarithmic factors) by a *critical radius* ε_n satisfying the relationship

$$\mathcal{H}(\mathcal{F}, \varepsilon_n) \asymp n\varepsilon_n^2, \tag{1}$$

where $\mathcal{H}$ denotes the *entropy* for the model class $\mathcal{F}$ under a suitable problem-specific notion of distance. Existing density estimation results have two limitations from the perspective of the present work. First, they are tailored to joint density estimation, and consequently require estimation of the marginal covariate distribution. This leads to suboptimal results for conditional density estimation where—as we show—estimating the marginal distribution is not required for optimal performance. Second, joint density estimation results typically rely on a *uniform* notion of entropy, which requires strong coverage over the entire covariate space; we show that for conditional density estimation, data-dependent ("empirical") notions of entropy are necessary to obtain tight rates, particularly when the covariate space is unbounded or high-dimensional. See Examples 1 and 2 and the discussion in Section 3.5 for more details.

Empirical entropy is known to characterize the minimax rates for various nonparametric regression settings [6, 65, 75]. However, the notion of distance (the L^p metric) used to define entropy in these settings is tailored to estimating conditional means, and may not be suitable for the more general problem of estimating conditional densities. In fact, recent work on a sequential variant of the conditional density estimation problem [9] shows that—in contrast to nonparametric regression—L^p entropy is not sufficient to characterize minimax rates for all conditional density classes simultaneously, even when responses are binary-valued. Consequently, determining the minimax rates for conditional density estimation with general conditional density classes has remained an open problem. The primary contribution of the present work is to provide a notion of entropy $\mathcal{H}(\mathcal{F}, \varepsilon_n)$ such that the minimax rates for conditional density estimation are governed by the relationship in equation (1) up to logarithmic factors.

Contributions. We characterize the minimax rates for conditional density estimation under Kullback–Leibler risk in terms of the *empirical Hellinger entropy* for the conditional density class. Our results establish matching (within logarithmic factors) upper and lower bounds on the KL risk for *all* conditional density classes with either *parametric* entropy ($\mathcal{H}(\mathcal{F}, \varepsilon_n)$ of size $p\log(1/\varepsilon_n)$) or *nonparametric* entropy ($\mathcal{H}(\mathcal{F}, \varepsilon_n)$ of size ε_n^{-p}). Our upper bounds are achieved using a novel estimator based on aggregation over an appropriate data-dependent discretization of $\mathcal{F}$ tailored to empirical Hellinger distance, and we complement the use of this estimator by highlighting natural settings in which the more standard maximum likelihood estimator attains suboptimal KL risk.

The key techniques that enable our results are (a) the novel introduction of empirical Hellinger entropy for analyzing conditional density estimation, (b) the use of concentration inequalities based on local Rademacher complexity, which is made possible by the aforementioned choice of empirical entropy, and (c) a result of Yang and Barron [81] that relates KL divergence to Hellinger distance whenever the *log* density ratio is bounded. The latter technique is what motivates our use of empirical Hellinger distance, and allows us to avoid requiring any lower bounds on the tails for the conditional density class under consideration. All three of these aspects are crucial to obtaining tight rates for general, potentially nonparametric classes of conditional densities.

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Organization. The remainder of this work is organized as follows. Section 2 gives the formal setup for the conditional density estimation problem. In Section 3, we state our main results; in particular, Section 3.3 describes an explicit estimator that achieves minimax rates, and Section 3.4 provides six examples for which the rates from our main theorem improve upon the state of the art. In Section 4, we state slightly more general variants of our primary theorems (upper and lower bounds for the minimax risk), and prove the main results. In Section 5, we discuss the performance of the conditional maximum likelihood estimator and prove that in general, it does not achieve minimax rates, and in Section 6 we highlight relevant related work from the rich literature on density estimation and nonparametric regression. Finally, we conclude with a brief discussion in Section 7. We defer omitted proofs for the main results, as well as a straightforward extension to obtain regret bounds and an adaptive estimator, to the Supplementary Material [10].

2. Problem setting. We consider a conditional density estimation setting in which the statistician receives i.i.d. samples $(x_1, y_1), \dots, (x_n, y_n) \in \mathcal{X} \times \mathcal{Y}$, where $x_t \sim \mu_{\mathcal{X}}$ is a covariate and $y_t | x_t \sim f^*(x_t)$ is a response, and where f^* is the true underlying conditional density function. Given such a sample, the goal of the statistician is to produce an estimator $\hat{f}_n$ such that the average KL divergence $\mathbb{E}_{x\sim\mu_{\mathcal{X}}} \mathrm{KL}(f^*(x) \,\|\, \hat{f}_n(x))$ is minimized. We next define the setting rigorously and state our main statistical assumptions.

2.1. *Probability spaces and statistical assumptions.* Let $(\mathcal{X}, \mathcal{A})$ and $(\mathcal{Y}, \mathcal{B})$ be measurable spaces denoting the covariate space and response space respectively. Our results are stated in terms of a fixed reference measure ν on $(\mathcal{Y}, \mathcal{B})$; let $\mathcal{M}(\mathcal{Y})$ denote the set of probability densities with respect to ν on $(\mathcal{Y}, \mathcal{B})$. A *(regular) conditional density* is a jointly measurable map $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ such that for all $x \in \mathcal{X}$, $[y \mapsto f(x, y)] \in \mathcal{M}(\mathcal{Y})$. We denote the collection of regular conditional densities by $\mathcal{K}(\mathcal{X}, \mathcal{Y})$. The *covariate distribution* $\mu_{\mathcal{X}}$ is an arbitrary probability measure on $(\mathcal{X}, \mathcal{A})$, and observations are generated by the process $x \sim \mu_{\mathcal{X}}$ and $y|x \sim f^*(x)$, where $f^* \in \mathcal{K}(\mathcal{X}, \mathcal{Y})$ is the true conditional density.

Let $\mathcal{F} \subseteq \mathcal{K}(\mathcal{X}, \mathcal{Y})$ be the statistician's *conditional density class* (or, *model*), and for each $x \in \mathcal{X}$, $y \in \mathcal{Y}$, and $f \in \mathcal{F}$, denote the conditional density of y given x by $[f(x)](y)$. For our results, we require that the model is *well specified* and bounded above.

ASSUMPTION 1. The true (data-generating) conditional density satisfies $f^* \in \mathcal{F}$.

ASSUMPTION 2. There exist constants $B, K < \infty$ such that $\sup_{f\in\mathcal{F}} \|f\|_\infty = B$ and $\nu(\mathcal{Y}) = K$. Without loss of generality we assume that $K \geq 1/B$.

While many of the quantities we consider (e.g., KL and Hellinger) are independent of the choice of reference measure ν, our quantitative results depend (weakly) on this choice through the parameters B and K. Refer to Section 3 for further discussion.

2.2. *Estimation and risk.* Let $\mathbb{Z}_+ = \mathbb{N} \cup \{0\}$, and for any $n \in \mathbb{N}$ let $[n] = \{1, \dots, n\}$. For any function f defined on an arbitrary space $\mathcal{C}$ and vector $c_{1:n} \in \mathcal{C}^n$, we use the vector notation $f(c_{1:n}) = (f(c_1), \dots, f(c_n))$. By a sample of size n, we mean a random element $S_n = (x_t, y_t)_{t\in[n]}$ taking values in $(\mathcal{X} \times \mathcal{Y})^n$. In this setting, an estimator (of f^*) is a sequence of regular-conditional-density-valued functions $\hat{f} = (\hat{f}_n)_{n\in\mathbb{Z}_+}$ where, for each n,

$$\hat{f}_n : (\mathcal{X} \times \mathcal{Y})^n \to \mathcal{K}(\mathcal{X}, \mathcal{Y}).$$

When the sample is clear from context, we use $\hat{f}_n$ to also denote the *estimate* $\hat{f}_n(S_n)$.

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For brevity, when x is a random element, we write $y \sim f^*(x)$ to denote $y|x \sim f^*(x)$. For any $n \geq 1$ and measurable $\phi : \mathcal{X}^n \to \mathbb{R}$, we write

$$\mathbb{E}_{x_{1:n} \sim \mu_{\mathcal{X}}}[\phi(x_{1:n})] = \int \phi(x_{1:n}) \mu_{\mathcal{X}}^{\otimes n}(\mathrm{d}x_{1:n}),$$

and similarly for any measurable $\psi : \mathcal{Y}^n \to \mathbb{R}$, $f^* \in \mathcal{K}(\mathcal{X}, \mathcal{Y})$, and $x_{1:n} \in \mathcal{X}^n$, we write

$$\mathbb{E}_{y_{1:n} \sim f^*(x_{1:n})}[\psi(y_{1:n})] = \int \psi(y_{1:n}) \left(\prod_{t=1}^{n} [f^*(x_t)](y_t) \right) \nu^{\otimes n}(\mathrm{d}y_{1:n}).$$

When $n = 1$ we drop the subscripts.

For any densities $p, q \in \mathcal{M}(\mathcal{Y})$, define the *Kullback–Leibler (KL) divergence* by

$$\mathrm{KL}(p \| q) = \int p(y) \log\left(\frac{p(y)}{q(y)} \right) \nu(\mathrm{d}y).$$

It will also be convenient to define the so-called *KL distance* by

$$d_{\mathrm{KL}}(p, q) = \sqrt{\mathrm{KL}(p \| q)}.$$

We define the *KL loss* of a regular conditional density $f \in \mathcal{K}(\mathcal{X}, \mathcal{Y})$ as

$$\mathcal{L}(f; f^*, \mu_{\mathcal{X}}) = \mathbb{E}_{x \sim \mu_{\mathcal{X}}} d_{\mathrm{KL}}^2(f^*(x), f(x)),$$

and define the *KL risk* of an estimator $\hat{f} = (\hat{f}_n)_{n \in \mathbb{N}}$ as

$$\mathcal{R}_n(\hat{f}; f^*, \mu_{\mathcal{X}}) = \mathbb{E}_{x_{1:n} \sim \mu_{\mathcal{X}}} \mathbb{E}_{y_{1:n} \sim f^*(x_{1:n})} \mathcal{L}(\hat{f}_n(S_n); f^*, \mu_{\mathcal{X}}).$$

Finally, the *minimax risk* of a conditional density class $\mathcal{F} \subseteq \mathcal{K}(\mathcal{X}, \mathcal{Y})$ is

$$\mathcal{R}_n(\mathcal{F}) = \inf_{\hat{f}} \sup_{\mu_{\mathcal{X}}} \sup_{f^* \in \mathcal{F}} \mathcal{R}_n(\hat{f}; f^*, \mu_{\mathcal{X}}),$$

where $\hat{f}$ ranges over all estimators and $\mu_{\mathcal{X}}$ ranges over all probability measures on $(\mathcal{X}, \mathcal{A})$. Our main contribution is tight upper and lower bounds on the minimax risk.

Additional notation. For any two densities $f, g \in \mathcal{M}(\mathcal{Y})$, we use the following standard discrepancy measures. For $r > 0$, the L^r *distance* is

$$d_r(f, g) = \left(\int |f(y) - g(y)|^r \nu(\mathrm{d}y) \right)^{1/r}$$

and the *Hellinger distance* is

$$d_{\mathrm{H}}(f, g) = \frac{1}{\sqrt{2}} d_2(\sqrt{f}, \sqrt{g}).$$

Notice that the Hellinger distance is scaled to ensure that $d_{\mathrm{H}} \in [0, 1]$.

If $\mathcal{Y} = \{0, 1\}$, then for any function $d : \mathcal{M}(\mathcal{Y}) \times \mathcal{M}(\mathcal{Y}) \to \mathbb{R}$ and $a, b \in [0, 1]$, we overload notation to let $d(a, b)$ denote $d(\mathrm{Ber}(a), \mathrm{Ber}(b))$.

Asymptotic notation. To state results concisely, we use standard asymptotic notation. In particular, for any two nonnegative sequences (a_n) and (b_n), we write $a_n \lesssim b_n$ if there exists a constant $C > 0$ such that for all n, $a_n \leq C b_n$. If $a_n \lesssim b_n$ and $b_n \lesssim a_n$, we write $a_n \asymp b_n$. All constants are independent of n, B, K, the effective dimension (i.e., p when $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n) \asymp \varepsilon^{-p}$), and the scale for the covering/packing numbers we consider.

3. Main results. In order to characterize the minimax risk $\mathcal{R}_n(\mathcal{F})$ of conditional density estimation, we must first formally introduce the *empirical Hellinger entropy*.

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3.1. *Empirical metric entropy.* Fix a metric $d : \mathcal{M}(\mathcal{Y}) \times \mathcal{M}(\mathcal{Y}) \to \mathbb{R}$ and normalized ℓ_q norm for $q \in [1, \infty)$. (The extension of the following definitions to $q = \infty$ is clear.) For a class $\mathcal{F} \subseteq \mathcal{K}(\mathcal{X}, \mathcal{Y})$, we say $\mathcal{G} \subseteq \mathcal{F}$ *(d, q)-covers $\mathcal{F}$ on $x_{1:n} \subseteq \mathcal{X}$ at scale $\varepsilon > 0$* if

$$\sup_{f \in \mathcal{F}} \inf_{g \in \mathcal{G}} \left(\frac{1}{n} \sum_{t=1}^{n} d^q\big(f(x_t), g(x_t)\big) \right)^{1/q} \leq \varepsilon.$$

We denote the cardinality of the smallest such cover by $\mathcal{N}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n})$. The *empirical (d, q)-entropy of $\mathcal{F}$ on samples of size n at scale ε* is

$$\mathcal{H}_{d,q}(\mathcal{F}, \varepsilon, n) = \sup_{x_{1:n} \subseteq \mathcal{X}} \log \mathcal{N}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n}).$$

Similarly, we say $\overline{\mathcal{G}} \subseteq \mathcal{F}$ is a *(d, q)-packing of $\mathcal{F}$ on $x_{1:n} \subseteq \mathcal{X}$ at scale $\varepsilon > 0$* if

$$\inf_{f \neq g \in \overline{\mathcal{G}}} \left(\frac{1}{n} \sum_{t=1}^{n} d^q\big(f(x_t), g(x_t)\big) \right)^{1/q} > \varepsilon.$$

We denote the cardinality of the largest such packing $\overline{\mathcal{G}}$ by $\overline{\mathcal{N}}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n})$. The *empirical (d, q)-packing entropy of $\mathcal{F}$ on samples of size n at scale ε* is

$$\overline{\mathcal{H}}_{d,q}(\mathcal{F}, \varepsilon, n) = \sup_{x_{1:n} \subseteq \mathcal{X}} \log \overline{\mathcal{N}}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n}).$$

These quantities are closely related, as shown by the following well-known result.

LEMMA 1. *For all $\varepsilon > 0$ and $x_{1:n} \subseteq \mathcal{X}$,*

$$\overline{\mathcal{N}}_{d,q}(\mathcal{F}, 2\varepsilon, x_{1:n}) \leq \mathcal{N}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n}) \leq \overline{\mathcal{N}}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n}).$$

Our main results, presented below, characterize the minimax rates for conditional density estimation in terms of the *empirical Hellinger entropy*, defined by

$$\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n) = \mathcal{H}_{d_{\mathrm{H}},2}(\mathcal{F}, \varepsilon, n).$$

That is, for the metric on densities in the general definition above we take $d = d_{\mathrm{H}}$, the Hellinger distance. Our use of the ℓ_q norm (in particular, ℓ_2) on the empirical observations is crucial for our results. For more detailed discussion of uniform alternatives and why they fail, see Section 3.5.

3.2. *Minimax upper and lower bounds.* To state our results in an easily interpretable fashion, we present results for classes $\mathcal{F}$ of either *parametric* or *nonparametric* complexity (for completeness, we also recover results for finite classes). Informally, we say that $\mathcal{F}$ is parametric if $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n)$ grows logarithmically in ε^{-1} and $\mathcal{F}$ is nonparametric if $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n)$ grows polynomially in ε^{-1}. This dichotomy is standard in the minimax literature, and captures most classes of interest.

Theorems 1 and 2 are simplifications of more general results, Theorems 3 and 4, which we state and prove in Section 4. Theorems 3 and 4 provide upper and lower bounds that hold for *any* class of conditional densities, with no assumption on the growth rate for the empirical entropy. Theorems 1 and 2 follow by applying these results with the parametric and nonparametric growth rates assumed in the theorem statements; see Appendix A of the Supplementary Material [10] for detailed calculations.

In all cases, we say a sequence r_n (which may depend on quantities such as p, B, and K) is *the minimax rate for $\mathcal{F}$* if $\mathcal{R}_n(\mathcal{F}) \asymp r_n$; as is standard in the literature, we often refer to

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the minimax rate as r_n even if this is only satisfied up to logarithmic factors. Our main results (Theorems 1 and 2) establish that, for general classes satisfying either the parametric or nonparametric growth assumptions above, the minimax KL risk is determined by the *critical radius* ε_n^2 satisfying (up to logarithmic factors)

$$\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon_n, n) \asymp n\varepsilon_n^2.$$

The precise dependence of ε_n on logarithmic factors will depend on $\mathcal{F}$; we discuss the necessity of these logarithmic factors in Section 3.5.

THEOREM 1 (Main upper bound). *For all* $p, C_{\mathcal{F},n} > 0$:
If $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n) \lesssim C_{\mathcal{F},n} \cdot \varepsilon^{-p}$ *and* $C_{\mathcal{F},n} \lesssim n^{\frac{p}{p+2}} \cdot \mathbb{I}\{p \le 2\} + n^{\frac{4}{p(p+2)}} \cdot \mathbb{I}\{p > 2\}$,

$$\mathcal{R}_n(\mathcal{F}) \lesssim C_{\mathcal{F},n}^{\frac{2}{p+2}} \cdot \left[\log(nBK)\right]^{\frac{p}{p+2}} \cdot n^{-\frac{2}{p+2}}.$$

If $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n) \lesssim p \cdot \log(C_{\mathcal{F},n}/\varepsilon)$ *and* $n \cdot C_{\mathcal{F},n}^2 > p$,

$$\mathcal{R}_n(\mathcal{F}) \lesssim \frac{p}{n} \cdot \left(\log(C_{\mathcal{F},n}\sqrt{n}) + \log(nBK) \cdot \log(C_{\mathcal{F},n}\sqrt{n}/\sqrt{p})\right) + \frac{\log(nBK) \cdot \log(n)}{n}.$$

THEOREM 2 (Main lower bound). *For all* $p, C_{\mathcal{F},n} > 0$ *and* n *sufficiently large*:
If $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n) \asymp C_{\mathcal{F},n} \cdot \varepsilon^{-p}$,

$$\mathcal{R}_n(\mathcal{F}) \gtrsim C_{\mathcal{F},n}^{\frac{2}{p+2}} \cdot \left[\log(nBK)\right]^{-\frac{2}{p+2}} \cdot n^{-\frac{2}{p+2}}.$$

If $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n) \asymp p \cdot \log(C_{\mathcal{F},n}/\varepsilon)$,

$$\mathcal{R}_n(\mathcal{F}) \gtrsim \frac{p}{n} \cdot \left(\frac{1}{\log(nBK)}\right).$$

REMARK 1. Since the Hellinger distance is upper bounded by KL, Theorem 1 immediately implies a bound on Hellinger risk. Similarly, inspection of the proof of Theorem 2 shows that we actually lower bound Hellinger risk, and hence our results imply that Hellinger and KL risk are equivalent up to logarithmic factors. It remains open to find a density class where they have differing dependence on logarithmic factors.

3.3. *Minimax estimator.* We now describe the estimator that achieves the minimax upper bound in Theorem 1 and provide some intuition behind its performance. The main difficulty in controlling performance under the KL risk is in handling the likelihood ratio, which a-priori is potentially unbounded. The obvious way to remove this problem is to require that all densities in $\mathcal{F}$ are bounded away from zero, but this would eliminate many classes of interest (e.g., shape-constrained densities). A first observation is that, under Assumption 2, we can restrict our estimator to be bounded away from zero without losing any generality in the conditional density class $\mathcal{F}$ via *smoothing*. In particular, for every $q \in \mathcal{M}(\mathcal{Y})$ and $\alpha > 0$, define the α*-smoothed density*

$$\widetilde{q}^{(\alpha)}(y) = \frac{q(y) + \alpha/K}{1+\alpha}, \quad y \in \mathcal{Y}.$$

Note that defining such a uniformly smoothed estimator necessitates the existence of a reference measure ν that admits finite K for $\mathcal{Y}$. However, this does not necessarily require that $\mathcal{Y}$ itself be bounded; for instance, in many standard settings it suffices to take ν to be another probability distribution with slightly heavier tails (see Example 1).

Our estimator is parametrized by scalars $\varepsilon, \alpha > 0$. Given a sample $(x_1, y_1), \dots, (x_{2n}, y_{2n})$, the estimator splits the sample into two halves $(x'_{1:n}, y'_{1:n})$ and $(x_{1:n}, y_{1:n})$. The main idea behind the estimator is to use the first half of the sample to form an empirical Hellinger cover, then use the second half of the sample to perform aggregation over a smoothing of the cover.

In more detail, let $\mathcal{G}$ be an $(\mathrm{H}, 2)$-cover of $\mathcal{F}$ on $x'_{1:n}$ at scale ε. Then, for each $t \in [n]$, define the mixture forecaster

$$[\hat{f}_t(x)](y) = \sum_{g \in \mathcal{G}} [\widetilde{g}^{(\alpha)}(x)](y) \left[\frac{\prod_{s=1}^{t} [\widetilde{g}^{(\alpha)}(x_s)](y_s)}{\sum_{h \in \mathcal{G}} \prod_{s=1}^{t} [\widetilde{h}^{(\alpha)}(x_s)](y_s)} \right], \quad x \in \mathcal{X}, y \in \mathcal{Y}.$$

The estimated conditional density is given by the Cesaro average of the mixture forecasters:

$$[\hat{f}_n^{\mathrm{agg}}(x)](y) = \frac{1}{n+1} \sum_{t=0}^{n} [\hat{f}_t(x)](y), \quad x \in \mathcal{X}, y \in \mathcal{Y}.$$

The weights of the mixture forecaster correspond to the posterior probability that each smoothed conditional density generated the observed data under a uniform prior. Such posteriors over a cover for the density class have previously been used to obtain minimax rates for joint density estimation (see, for example, [80]). Crucially for this work, however, our estimator uses the posterior over an *empirical cover* that must be estimated from data, and we explicitly smooth the estimator in order to avoid unbounded likelihood ratios.

3.4. *Examples*. We now instantiate Theorems 1 and 2 for concrete function classes of interest. The main contribution of this section is to provide the minimax rates for three models that existing joint density estimation bounds *fail* to capture. Our examples include a high-dimensional covariate space, an unbounded covariate space, and multiple conditional analogues of smoothness classes. In each example, we obtain upper bounds using Theorem 1 (and, in most cases, matching lower bounds using Theorem 2), and the rates are achieved by the minimax estimator introduced in Section 3.3. Proofs for examples are deferred to Appendix B of the Supplementary Material [10].

To instantiate our main theorem, all that is required is to compute the empirical Hellinger entropy for the conditional density class of interest. While the empirical L^2 entropy for regression functions has been extensively studied, empirical Hellinger entropy cannot be tightly bounded by L^r entropy for any r. This is because the tightest generic inequality possible for the Hellinger distance is

$$d_1 \le d_{\mathrm{H}} \le \sqrt{d_1}, \tag{2}$$

and which side of the inequality is tight will depend on the geometry of the densities. In our examples, for some classes (e.g., smooth densities) we find that $\mathcal{N}_{\mathrm{H}} \asymp \mathcal{N}_{\|\cdot\|_1^{1/2}}$, while for others (e.g., linear functions) we find that $\mathcal{N}_{\mathrm{H}} \asymp \mathcal{N}_{\|\cdot\|_1}$. In the context of joint density estimation, Birgé [12] makes a similar observation, noting that this adaptive behavior is what allows d_{H} to correctly capture the complexity of (joint) density estimation. This contrasts with the problem of conditional mean estimation under square loss, where L^2 entropy is sufficient to completely characterize the minimax rates [65].

Dimension-free generalized linear models. Consider univariate generalized linear models (GLMs) with exponential family responses and canonical link functions (cf. [59]). Classical asymptotic rates for consistency and normality are well understood for GLMs, where the maximum likelihood error scales with the Fisher information [32]. We now use our main results to extend the study of GLMs beyond this classical setting.

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Parameter estimation for GLMs is *exactly* a regression problem, yet in this work we are interested in a full characterization of the uncertainty of the response variable through the conditional density. Hence, we depart from the classical theory by studying the KL error of the estimated conditional density rather than error on the parameter scale, which *a priori* may be unbounded even for bounded parameters. For exponential families satisfying certain tail conditions (e.g., logistic regression with probabilities restricted to some fixed interval $[c, 1-c]$ for $c > 0$), earlier work [26, 40, 82] obtains the parametric KL rate $p \cdot (\log n) \cdot n^{-1}$ (where p is the covariate dimension). Without any tail assumptions (although we generally require bounds on the scale parameter to ensure the density is bounded above), our Theorem 1 recovers this rate up to logarithmic factors for generic exponential family responses, since the entropy is parametric.

In the regime where the parameter dimension scales with n, this parametric rate is no longer useful. Under a sparsity assumption that only k entries of the linear regressor are nonzero, it has been shown in a variety of cases that minimax risk of $k \cdot (\log p) \cdot n^{-1}$ is possible [66, 75]. In the case of a dense linear predictor or exponentially large dimension, these rates become vacuous. In the linear regression setting—where similar parametric dependence on dimension arises—it is known that, in fact, dimension-free rates are possible [4]. Finally, for Bernoulli responses with a linear link function, Lemma 8 of Rakhlin and Sridharan [63] (combined with Theorem D.2 of the Supplementary Material [10]) implies that the minimax risk is no larger than $\sqrt{(\log n) \cdot n^{-1}}$. Our contribution in this section is to show that the same dimension-free rates are possible for the KL risk of conditional density estimation of generalized linear models on $\mathbb{R}$.

EXAMPLE 1. Let $\mathbb{H}$ be a Hilbert space, and for any radius $R > 0$ let $\mathcal{B}(R) = \{v \in \mathbb{H} | \|v\| \le R\}$. For each of the following, the polynomial dependence on n is tight.

(a) Fix $\sigma, X, W > 0$. Let $\mathcal{Y} = \mathbb{R}$ and set $\mathcal{X} = \mathcal{B}(X)$ and $\mathcal{W} = \mathcal{B}(W)$. If

$$\mathcal{F}_{\mathcal{W}} = \{x \mapsto \text{Gaussian}(\langle x, w\rangle, \sigma^2) | w \in \mathcal{W}\},$$

then

$$\mathcal{R}_n(\mathcal{F}_{\mathcal{W}}) \lesssim n^{-1/2} \cdot \frac{XW \cdot \sqrt{\log n} \cdot \sqrt{\log(nX^2W^2/\sigma)}}{\sigma}.$$

(b) Fix $X, W > 0$. Let $\mathcal{Y} = \mathbb{N}$ and set $\mathcal{X} = \mathcal{B}(X)$ and $\mathcal{W} = \mathcal{B}(W)$. If

$$\mathcal{F}_{\mathcal{W}} = \{x \mapsto \text{Pois}(\exp\{\langle x, w\rangle\}) | w \in \mathcal{W}\},$$

then

$$\mathcal{R}_n(\mathcal{F}_{\mathcal{W}}) \lesssim n^{-1/2} \cdot XWe^{XW/2} \cdot (\log n).$$

(c) Fix $\alpha, \gamma, X, W > 0$. Let $\mathcal{Y} = [0, \infty)$ and set $\mathcal{X} = \mathcal{B}(X)$ and $\mathcal{W} = \mathcal{B}(W)$. If

$$\mathcal{F}_{\mathcal{W}} = \left\{x \mapsto \text{Gamma}\left(\alpha, -\frac{1}{\alpha(\langle x, w\rangle - XW - \gamma)}\right) \Big| w \in \mathcal{W}\right\},$$

then

$$\mathcal{R}_n(\mathcal{F}_{\mathcal{W}}) \lesssim n^{-1/2} \cdot (\log n) \cdot \sqrt{\frac{XW \max\{\alpha, 1\}}{\gamma} \log\left(n \cdot \left(\frac{XW + \gamma}{\gamma}\right)^{\alpha}\right)}.$$

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Vapnik–Chervonenkis (VC) classes. While the above example has removed all dependence on the dimension of the covariate space, the rate still depends on the norm of the linear regressor, which is generally unavoidable without assumptions on $\mu_{\mathcal{X}}$ [56]. However, if the conditional density map is further restricted, we now show that it is possible to consider unbounded covariate spaces without any assumptions on $\mu_{\mathcal{X}}$.

For the next example, we consider a classification setting where $\mathcal{Y} = \{0, 1\}$, so that any density in $\mathcal{M}(\mathcal{Y})$ can represented by a scalar in $[0, 1]$ parameterizing the mean of a Bernoulli random variable. Taking ν to be uniform on $\{0, 1\}$ gives $K = 1$ and $B = 2$. This setting has been extensively studied, particularly in the sequential, non-i.i.d. setting (see Section 6.3 for a survey), and is closely connected to the problem of universal compression. A common measure of complexity for classification is the *VC-dimension*, and the natural extension to real-valued functions is the *pseudo-dimension*. Our example concerns the case where the conditional density class has a finite pseudo-dimension; for completeness, we first restate the relevant definitions.

Let $\mathcal{P}(\mathcal{X})$ denote the power set of $\mathcal{X}$. For any $\mathcal{C} \subseteq \mathcal{P}(\mathcal{X})$, its *VC-dimension* is

$$\mathrm{VCdim}(\mathcal{C}) = \sup\bigl\{n \in \mathbb{N} \mid \exists x_{1:n} \subseteq \mathcal{X} \text{ s.t. } \bigl|\{c \cap x_{1:n} \mid c \in \mathcal{C}\}\bigr| = 2^n\bigr\}.$$

For any $\mathcal{F} \subseteq \mathcal{X} \to \mathbb{R}$, its *pseudo-dimension* is

$$\mathrm{Pdim}(\mathcal{F}) = \mathrm{VCdim}\bigl(\bigl\{\{(x, v) \in \mathcal{X} \times \mathbb{R} \mid f(x) \geq v\} \mid f \in \mathcal{F}\bigr\}\bigr).$$

Then, we have the following result.

EXAMPLE 2. If $\mathcal{X}$ is arbitrary, $\mathcal{Y} = \{0, 1\}$, and $\mathcal{F} \subseteq \mathcal{X} \to [0, 1]$, then

$$\mathcal{R}_n(\mathcal{F}) \lesssim \frac{\mathrm{Pdim}(\mathcal{F})(\log n)^2}{n}.$$

Bhatt and Kim [8] consider a special case of this problem in the sequential (but still i.i.d.) setting when $\mathcal{F} = \{x \mapsto \theta_{\mathbb{I}\{x \in c\}} \mid c \in \mathcal{C}, \theta_0, \theta_1 \in [0, 1]\}$ for some fixed $\mathcal{C} \subseteq \mathcal{P}(\mathcal{X})$. They provide a risk upper bound $\mathcal{R}_n(\mathcal{F}) \lesssim \sqrt{\mathrm{VCdim}(\mathcal{C}) \cdot n^{-1}}$ (see Theorem D.2 of the Supplementary Material [10] to relate their result—which is stated in terms of regret—to a risk bound), which has a gap from their lower bound, $\mathrm{VCdim}(\mathcal{C}) \cdot n^{-1}$. To resolve this gap, we require the following lemma, which we also prove in Appendix B of the Supplementary Material [10].

LEMMA 2. *For any $\mathcal{C} \subseteq \mathcal{P}(\mathcal{X})$, if $\mathcal{F} = \{x \mapsto \theta_{\mathbb{I}\{x \in c\}} \mid c \in \mathcal{C}, \theta_0, \theta_1 \in [0, 1]\}$ then*

$$\mathrm{Pdim}(\mathcal{F}) \leq 6 \cdot \mathrm{VCdim}(\mathcal{C}).$$

Combining Example 2 and Lemma 2, our minimax estimator achieves an upper bound that significantly improves on the $n^{-1/2}$ rate, matching the lower bound of Bhatt and Kim [8] within $\log(n)$ factors.

So-called "fast rates" of the form $\mathrm{Pdim}(\mathcal{F}) \cdot n^{-1}$ for VC classes have been extensively studied in the literature. For binary outcomes (i.e., Example 2), constraining the probabilities away from the boundary of $[0, 1]$ ensures that the logarithmic loss is bounded above and below by square loss (similarly for Hellinger distance or total variation), and hence classical results for regression apply (e.g., Theorem 8 of [45]). Without such boundedness assumptions, the discussion in Sections 2.3 and 7 of Yang and Barron [80] implies fast rates for the KL risk of joint density estimation. However, even classes with $\mathrm{VCdim}(\mathcal{C}) = 1$ may not admit a uniform cover (e.g., halfspaces on $\mathbb{R}$), and consequently our use of empirical entropy is crucial to obtain nonvacuous rates. Further, it is possible to have $\mathrm{VCdim}(\mathcal{C}) = 1$ yet fail to be *online learnable* (again, one-dimensional halfspaces witness this; see Example 21.4 of [7]), in which case the upper bounds provided by worst-case sequential analyses (e.g., Theorem 7 of [33]) will be vacuous. For the KL risk of conditional density estimation, Zhang [82] and Grünwald and Mehta [40] obtain fast rates under boundedness assumptions on the densities. While these assumptions are weaker than strictly constraining the densities away from the boundary of [0, 1], they fail for simple classes (including the $\mathcal{F}$ in Lemma 2). In summary, to the best of our knowledge, Example 2 provides the first fast rates for KL risk of VC conditional density classes without any boundedness assumptions.

Smoothness classes. Smooth densities are one of the most fundamental classes studied in joint density estimation, and their entropy is well understood (e.g., Theorem 3 of [47]). For conditional densities, there are multiple possible notions of smoothness, and these have been studied on a case-by-case basis in the literature with a new (kernel-based) estimator for each setting. We now consider three possible notions, providing novel rates for the KL risk in each case.

First, consider the case when the response space is discrete; that is, conditional multinomial estimation.

EXAMPLE 3. Let $\mathcal{X} = [0, 1]^p$, $\mathcal{Y} = [k]$, $\gamma \in (0, 1]$, and $m \in \mathbb{Z}_+$ such that

$$\mathcal{F} = \left\{ x \mapsto \big(\psi_1(x), \dots, \psi_k(x)\big) \,\middle|\, \begin{array}{l} \forall x, x' \in \mathcal{X},\ j \in [k], \quad \sup\limits_{\alpha:\|\alpha\|_1 \le m} \big|D^\alpha \psi_j(x)\big| \le 1 \text{ and} \\ \sup\limits_{\alpha:\|\alpha\|_1 = m} \big|D^\alpha \psi_j(x) - D^\alpha \psi_j(x')\big| \le \|x - x'\|_\infty^\gamma \end{array} \right\}.$$

Then, up to logarithmic factors, $\mathcal{R}_n(\mathcal{F}) \asymp k^{\frac{2p}{p+m+\gamma}} \cdot n^{-\frac{m+\gamma}{p+m+\gamma}}$.

In this setting with $\gamma = 1$ and $m = 0$, Theorem 2 of Györfi and Kohler [41] shows that the total variation risk is no larger than $n^{-\frac{1}{p+2}}$. The minimax KL rate of Example 3, which is $n^{-\frac{1}{p+1}}$, cannot be recovered from this result (and similarly the total variation risk cannot be recovered from our rate) using either of the inequalities in equation (2). Once again, this demonstrates how the KL risk is qualitatively different due to the curvature of the loss near the boundary.

For a continuous response space, more structure is required. Theorem 2.2 of Li, Neykov and Balakrishnan [57] shows that, in general, the conditional density must be smooth in *both* the covariates and the responses in order for the model to be estimable. One way to enforce this is to consider sufficiently smooth parametric densities with the parameters themselves generated by smooth maps; a concrete example of such parametric densities are exponential family models (see the proofs of Example 1). In particular, we have the following result, which follows immediately from Example 3.

EXAMPLE 4. Let $\mathcal{X} = [0, 1]^p$, $\mathcal{Y} = \mathbb{R}$, $\Theta > 0$, and $\gamma \in (0, 1]$. Suppose $\mathcal{P} = \{\mathcal{P}_\theta | \theta \in [0, \Theta]\}$ satisfies $d_{\mathrm{H}}^2(\mathcal{P}_\theta, \mathcal{P}_{\theta'}) \le |\theta - \theta'|$ for all $\theta, \theta' \in [0, \Theta]$, and define

$$\mathcal{F}_{\mathcal{P}} = \left\{ x \mapsto \mathcal{P}_{\psi(x)} | \forall x, x' \in \mathcal{X}, \psi(x) \in [0, \Theta] \text{ and } \big|\psi(x) - \psi(x')\big| \le \|x - x'\|_\infty^\gamma \right\}.$$

Then, up to logarithmic factors, $\mathcal{R}_n(\mathcal{F}) \lesssim \Theta^{\frac{2p}{p+\gamma}} \cdot n^{-\frac{\gamma}{p+\gamma}}$.

More generally, using recent *functional* metric entropy results from Park and Muandet [62], we obtain upper bounds (packing numbers, and hence lower bounds, remain an open problem) on the risk for models with smoothness constraints on both the covariates and the responses.

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EXAMPLE 5. Fix $\mathcal{X} = [0,1]^p$, $\mathcal{Y} = [0,1]^d$, $\gamma \in (0,1]$, $C, L > 0$, and $m, m' \in \mathbb{Z}_+$. If

$$\mathcal{F} = \left\{ x \mapsto \big(y \mapsto [f(x)](y)\big) \;\middle|\; \begin{array}{l} \forall x \in \mathcal{X}, y, y' \in \mathcal{Y}, \alpha \in \{0, \dots, m\}^p \text{ s.t. } \|\alpha\|_1 \le m, \\ \alpha' \in \{0, \dots, m'\}^d \text{ s.t. } \|\alpha'\|_1 \le m', \\ \big|D^{\alpha'}[D^\alpha f(x)](y)\big| \le C \text{ and} \\ \big|D^{\alpha'}[D^\alpha f(x)](y) - D^{\alpha'}[D^\alpha f(x)](y')\big| \le L\|y - y'\|_\infty^\gamma \end{array} \right\},$$

then up to factors in L and C and logarithmic factors, $\mathcal{R}_n(\mathcal{F}) \lesssim n^{-\frac{1}{1+p/m+d/(m'+\gamma)}}$.

Conditional mean estimation. In the context of nonparametric regression, Rakhlin, Sridharan and Tsybakov [65] characterized the minimax rates for conditional mean estimation with square loss, showing that if $\mathcal{H}_{2,2}(\mathcal{F}, \varepsilon, n) \asymp \varepsilon^{-p}$, the minimax rate for L^2 risk has $\mathbb{E}(f^*(x) - \hat{f}_n(x))^2 \asymp n^{-2/(2+p)}$, where $\mathcal{F} \subseteq \mathcal{X} \to \mathcal{Y}$ is a class of conditional *means*. In particular, their Theorem 2 provides an upper bound that holds for every function class and noise distribution (with $\mathcal{Y} \subseteq [0,1]$), while their Theorem 7 provides a specific function class and noise distribution ($\mathcal{Y} = \{0,1\}$) for which the lower bound holds. It appears to be folklore that their Theorem 7 can be upgraded to show that for *every* function class there exists a noise distribution such that the lower bound holds. Here we derive this result as a corollary of our lower bound to facilitate ease of use for both experts and nonexperts.

EXAMPLE 6. Let $\mathcal{X}$ be arbitrary, $\mathcal{Y} = \mathbb{R}$, and $\mathcal{F} \subseteq (\mathcal{X} \to [0,1])$ satisfy $\mathcal{H}_{2,2}(\mathcal{F}, \varepsilon, n) \asymp \varepsilon^{-p}$ for some $p > 0$. Then,

$$\inf_{\hat{\theta}} \sup_{\mu_{\mathcal{X}}} \sup_{f^* \in \mathcal{F}} \mathbb{E}_{x_{1:n} \sim \mu_{\mathcal{X}}} \mathbb{E}_{y_{1:n} \sim \mathrm{Gaussian}(f^*(x_{1:n}),1)} \mathbb{E}_{x \sim \mu_{\mathcal{X}}} \big(f^*(x) - \hat{\theta}_n(x)\big)^2 \gtrsim n^{-\frac{2}{2+p}}$$

up to logarithmic factors, where $\hat{\theta}$ ranges over conditional mean estimators.

3.5. *Discussion*. We close this section by discussing some of the key technical features of Theorems 1 and 2, as well as limitations and opportunities for improvement.

On logarithmic factors. Theorems 1 and 2 combined characterize the polynomial dependence on n for *all* function classes with parametric or nonparametric empirical Hellinger entropy. For marginal density estimation (no covariates), the dependence on logarithmic factors is unresolved for general classes; for example, the logarithmic factors do not match in Yang and Barron [80], Zhang [82], or Rigollet [67], and are ignored in more recent work such as Grünwald and Mehta [40]. Further, information-theoretic lower bounds are generally suboptimal with regards to logarithmic factors, which is a consequence of them holding for *all* function classes simultaneously rather than a specific function class constructed for a counterexample. For a related discussion of logarithmic factors in risk bounds, we direct the reader to Section 6.2.2 of Birgé and Massart [16]. We now briefly discuss more concretely the tightness of logarithmic factors for our setting.

When $\mathcal{F}$ is an arbitrary finite class, our analysis recovers $\mathcal{R}_n(\mathcal{F}) \lesssim (\log n) \cdot \log(nBK)/n$ (follows from part (iii) of Lemma 5), yet the correct rate is well known to be $(\log|\mathcal{F}|)/n$. This can be obtained with no additional assumptions and with no dependence on B and K by using, for example, Vovk's aggregating algorithm. A simple proof can be found by combining online-to-batch (Theorem D.2 of the Supplementary Material [10]) with the argument in Section 9.5 of Cesa-Bianchi and Lugosi [22], and a matching lower bound is provided by Theorem 3 of Lecué [55]. The looseness of our rate for finite classes is a consequence of our method and analysis applying to *every* function class simultaneously.

Beyond finite classes, it is impossible for the logarithmic factors to be tight in all cases without further assumptions beyond entropy conditions. Gaussian mean estimation with a

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known variance has a known minimax rate of $1/n$ (no logarithmic factors), and has entropy that grows like $\log(1/\varepsilon)$. However, Example 6 of Rakhlin, Sridharan and Tsybakov [65]—which applies in our setting since KL is lower bounded by square loss for Bernoulli conditional mean estimation—provides a function class that also has entropy that grows like $\log(1/\varepsilon)$, yet has minimax risk lower bounded by $(\log n)/n$.

We now demonstrate that our dependence on logarithmic factors can also not be improved in general without further assumptions, since they are tight for *some* function class. As a consequence, we conclude that while Hellinger empirical entropy correctly captures the polynomial dependence on n for the minimax rates, it is insufficient to fully characterize the logarithmic factors. It remains an open question to provide a more refined characterization of function classes that does capture this dependence.

FACT 1. There exists an $\mathcal{F}$ for which the dependence on B and K is tight.

PROOF OF FACT 1. We realize this lower bound for the unconditional density estimation setting, which is a subset of conditional density estimation. In particular, let $\mathcal{X} = \varnothing$, ν be Lebesgue measure, and $\mathcal{F}$ be all monotone densities on $[0, K]$ bounded above by B. Since metric entropy is no larger than *bracketing* entropy for any metric, Theorem 3 of Gao and Wellner [35] (instantiated with $k = 1$) implies that

$$\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n) \lesssim \sqrt{\log(BK)} \cdot \varepsilon^{-1}.$$

Thus, Theorem 1 implies that

$$\mathcal{R}_n(\mathcal{F}) \lesssim \big[\log(BK)\big]^{1/3} \cdot \big[\log(nBK)\big]^{1/3} \cdot n^{-2/3}.$$

However, by Pinsker's and equation (1.3) of Birgé [13],

$$\mathcal{R}_n(\mathcal{F}) \gtrsim \big[\log(BK)\big]^{2/3} \cdot n^{-2/3}. \qquad \square$$

FACT 2. There exists an $\mathcal{F}$ for which the dependence on $\log(n)$ is tight.

PROOF OF FACT 2. We again consider a collection of monotone densities, but this time in the regime where the support scales with n. In particular, let $\mathcal{X} = \varnothing$, ν be counting measure, and $\mathcal{F}$ be all monotone probability mass functions on $[K]$ bounded above by B. Taking $B = 1$ and $K = 2n^{1/3}$, the same argument as Fact 1 implies

$$\mathcal{R}_n(\mathcal{F}) \lesssim \left(\frac{\log n}{n}\right)^{2/3}.$$

However, by Pinsker's and Theorem 2.1 of Devroye and Reddad [28],

$$\mathcal{R}_n(\mathcal{F}) \gtrsim \left(\frac{\log n}{n}\right)^{2/3}. \qquad \square$$

Finally, for extremely rich classes with $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n)$ that grows exponentially in ε^{-1}, our results imply that the risk decays polynomially in $(\log n)^{-1}$. A more refined analysis is needed to understand the precise role of the logarithmic factors, which become crucially important when all dependence on n is logarithmic.

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Comparison with uniform joint entropy. As discussed in the Introduction, a key aspect that distinguishes our results from the classical literature on density estimation is that we work with the empirical Hellinger entropy of the conditional density class, which (i) is data-dependent, and (ii) captures only the difficulty of estimating the conditional distribution over responses, and *not* the difficulty of estimating the marginal distribution over covariates. To illustrate this improvement, we now highlight that (a) without assumptions on the marginal distribution of the covariates, which is the setting we study, uniform joint entropy leads to vacuous rates, and (b) even under some standard assumptions on the marginal distribution of the covariates, which is an easier problem than the one we consider, uniform joint entropy leads to suboptimal rates.

Let us formally define uniform joint entropy. Suppose there exists a finite measure λ on $\mathcal{X} \times \mathcal{Y}$, and let $\mathcal{M}(\mathcal{X} \times \mathcal{Y})$ denote the collection of densities on $\mathcal{X} \times \mathcal{Y}$ with respect to λ. For a class of densities $\Pi \subseteq \mathcal{M}(\mathcal{X} \times \mathcal{Y})$, we say $\Pi' \subseteq \Pi$ is a *uniform Hellinger cover* at scale $\varepsilon > 0$ if

$$\sup_{\pi \in \Pi} \inf_{\pi' \in \Pi'} d_{\mathrm{H}}(\pi, \pi') \le \varepsilon,$$

where the Hellinger distance for joint distributions is given by

$$d_{\mathrm{H}}(\pi, \pi') = \sqrt{\frac{1}{2} \int \left(\sqrt{\pi(x, y)} - \sqrt{\pi'(x, y)}\right)^2 \lambda\big(\mathrm{d}(x, y)\big)}.$$

We denote the cardinality of the smallest Π' that achieves this by $\mathcal{N}_{\mathrm{H}}(\Pi, \varepsilon)$. Existing minimax results for density estimation [78, 80] bound the minimax risk of estimating the data-generating density $\pi^* \in \Pi$ in terms of the *uniform joint entropy* $\mathcal{H}_{\mathrm{H}}(\Pi, \varepsilon) = \log \mathcal{N}_{\mathrm{H}}(\Pi, \varepsilon)$.

To compare the empirical conditional entropy to the uniform joint entropy, consider a set $\mathscr{D}$ of marginal densities on $\mathcal{X}$ and class $\mathcal{F} \subseteq \mathcal{K}(\mathcal{X}, \mathcal{Y})$, and define

$$\Pi_{\mathscr{D},\mathcal{F}} = \big\{(x, y) \mapsto \pi_{p,f}(x, y) = p(x)\big[f(x)\big](y) | p \in \mathscr{D}, f \in \mathcal{F}\big\}.$$

For any $p, p' \in \mathscr{D}$ and $f \in \mathcal{F}$, $d_{\mathrm{H}}(\pi_{p,f}, \pi_{p',f}) = d_{\mathrm{H}}(p, p')$ since $y \mapsto [f(x)](y)$ integrates to one for almost all x. Thus, $\mathcal{H}_{\mathrm{H}}(\Pi_{\mathscr{D},\{f\}}, \varepsilon) = \mathcal{H}_{\mathrm{H}}(\mathscr{D}, \varepsilon)$ for all $f \in \mathcal{F}$, so by monotonicity of metric entropy we conclude $\mathcal{H}_{\mathrm{H}}(\Pi_{\mathscr{D},\mathcal{F}}, \varepsilon) \ge \mathcal{H}_{\mathrm{H}}(\mathscr{D}, 2\varepsilon)$ for all $\mathcal{F}$. Trivially, if $\mathcal{H}_{\mathrm{H}}(\mathscr{D}, 2\varepsilon) = \infty$, then $\mathcal{H}_{\mathrm{H}}(\Pi_{\mathscr{D},\mathcal{F}}, \varepsilon) = \infty$. This is the case when, for example, $\mathcal{X} = \mathbb{R}$ and $\mathscr{D}$ is the set of all densities with respect to Lebesgue measure. In contrast, our results provide tight rates for this same setting.

The previous example has already ruled out the suitability of uniform joint entropy for conditional density estimation. However, even if $\mathcal{H}_{\mathrm{H}}(\Pi_{\mathscr{D},\mathcal{F}}, \varepsilon) < \infty$, uniform joint entropy may still lead to suboptimal rates under "natural" assumptions on $\mathscr{D}$. For example, if $\mathcal{X}$ is a compact, convex subset of $\mathbb{R}^d$, the marginal entropy of $\mathscr{D}$ grows as ε^{-d} for Lipschitz densities [30] or as ε^{-d-1} for almost-isotropic log-concave densities [52]. Since we generally expect that the dimension of $\mathcal{X}$ is large compared to the dimension of $\mathcal{Y}$ (commonly 1, as in regression and classification), this dependence is prohibitively suboptimal even when $\mathcal{F}$ has moderate nonparametric complexity. By characterizing performance in terms of the entropy for the conditional densities, our results only depend on the complexity of $\mathcal{X}$ through the complexity of $\mathcal{F}$ itself. For example, if $\mathcal{F}$ is parametric, our dependence on the dimension of $\mathcal{X}$ will be linear at worst, while the previous joint examples will have an exponential dependence.

These examples demonstrate why the entropy for the conditional density class $\mathcal{F}$ is the correct object to consider in our setting, rather than the joint entropy of $\Pi_{\mathscr{D},\mathcal{F}}$. If one does not use the *empirical* entropy for $\mathcal{F}$, however, one may obtain suboptimal rates. For example, a commonly used alternative notion of covering is to say that $\mathcal{G}$ uniformly covers $\mathcal{F}$ at scale ε if

$$\sup_{f\in\mathcal{F}}\inf_{g\in\mathcal{G}}\sup_{x\in\mathcal{X}} d_{\mathrm{H}}\big(f(x), g(x)\big) \le \varepsilon.$$

For large (continuous) $\mathcal{X}$, this may be vacuously large; this phenomenon is well known within the literature on regression with the square loss [58].

Finally, we briefly remark that if the covariate density $p_{\mu_{\mathcal{X}}}$ is known in advance, the conditional density estimation problem reduces to joint density estimation over the class $\{(x, y) \mapsto p_{\mu_{\mathcal{X}}}(x)[f(x)](y) | f \in \mathcal{F}\}$. However, the covariate density is rarely known in practice, which necessitates the use of empirical entropy.

On the bounded density assumption. To handle all classes simultaneously, we require the conditional densities to be bounded above, since otherwise (without additional assumptions) the KL risk may be infinite. For certain classes (e.g., exponential families), a more specialized analysis could likely replace this assumption with weaker tail conditions, or remove it altogether. Crucially, unlike related work [40, 82] that obtains bounds for classes satisfying certain tail conditions on the logarithmic loss composed with the density class, we do not require the true conditional density to be bounded below, which would rule out many classes of interest, particularly for classification applications. For a more detailed discussion, see Section 6.4.

The requirement that the reference distribution ν over $\mathcal{Y}$ admits a finite bound K appears as an artifact of the analysis of our minimax estimator (Section 3.3). While such an assumption is common in the minimax estimation literature, it is unclear whether it is necessary to obtain rates for all classes simultaneously. Ultimately, however, the logarithmic dependence on B and K in our results is fairly mild.

In more detail, observe that in many real statistical applications, the response space $\mathcal{Y}$ is often fairly simple. At the same time, the covariate space $\mathcal{X}$ can be anything but simple. For classification (finite $\mathcal{Y}$), $B = 1$ and $K = |\mathcal{Y}|$, and consequently the dependence on $\mathcal{Y}$ in our bound is only of size $\log|\mathcal{Y}|$. Another well-studied setting is where $\mathcal{Y}$ is a ball of radius r in d dimensions, in which case $K = r^d$, so the dependence on $\mathcal{Y}$ in our bound is $d \log r$. For the commonly encountered situation in which $\mathcal{X} \subseteq \mathbb{R}^p$ for some $d \ll p$, this dependence on d is of lower order. (Specifically for Example 1, one can take $d = 1$ and $r = K$.)

A natural choice for ν is the uniform distribution, if such a distribution exists. When no such distribution exists or such a choice does not permit a bound on the densities, one may design ν for the model class of interest (e.g., by picking ν to have slightly heavier tails). See Example 1 for specific realizations of ν that admit unbounded $\mathcal{Y}$.

4. Proofs for main theorems. We now prove Theorems 1 and 2. Before proceeding to the proofs, we introduce some additional technical tools.

4.1. *Technical preliminaries.*

Hellinger distance. We begin by stating two technical lemmas that allow one to relate the KL divergence to the Hellinger distance, which are crucial to our arguments. The first lemma shows that the two divergences are equivalent up to logarithmic factors whenever the density ratio for the arguments is bounded.

LEMMA 3 (Lemma 4 of Yang and Barron [81]). *For all $p, q \in \mathcal{M}(\mathcal{Y})$,*

$$d_{\mathrm{KL}}^2(p, q) \le 2\left[2 + \sup_{y\in\mathcal{Y}} \log\left(\frac{p(y)}{q(y)}\right)\right] d_{\mathrm{H}}^2(p, q).$$

The second lemma controls the approximation error incurred by the smoothing used in our estimator (which controls the aforementioned density ratio).

LEMMA 4. *For all* $\alpha \geq 0$ *and* $p, q \in \mathcal{M}(\mathcal{Y})$,

$$d_{\mathrm{H}}^2(p, \widetilde{q}^{(\alpha)}) \leq d_{\mathrm{H}}^2(p, q) + \alpha.$$

PROOF OF LEMMA 4. By convexity of squared Hellinger distance,

$$d_{\mathrm{H}}^2(p, \widetilde{q}^{(\alpha)}) \leq \frac{1}{1+\alpha} d_{\mathrm{H}}^2(p, q) + \frac{\alpha}{1+\alpha} d_{\mathrm{H}}^2(p, 1/K) \leq d_{\mathrm{H}}^2(p, q) + \alpha. \qquad \square$$

Local Rademacher complexity. An important step in the analysis for our upper bound is to derive "fast" (i.e., $1/n$-type for parametric classes) uniform concentration guarantees for Hellinger distance, which we obtain using local Rademacher complexities [5, 18, 51].

We define the *(empirical) Rademacher complexity* for a class $\mathcal{J} \subseteq \mathcal{X} \to \mathbb{R}$ on a sample $x_{1:n} \subseteq \mathcal{X}$ as

$$\mathfrak{R}_n(\mathcal{J}, x_{1:n}) = \mathbb{E}_{\sigma_{1:n}} \sup_{h \in \mathcal{J}} \frac{1}{n} \sum_{t=1}^{n} \sigma_t h(x_t),$$

where each σ_t is an independent Rademacher random variable, taking values 1 and -1 with equal probability. The *worst-case Rademacher complexity* over samples is given by

$$\mathfrak{R}_n(\mathcal{J}) = \sup_{x_{1:n}} \mathfrak{R}_n(\mathcal{J}, x_{1:n}).$$

The *localization* of $\mathcal{J}$ at scale $r > 0$ on a sample $x_{1:n}$ is defined as

$$\mathcal{J}[r, x_{1:n}] = \left\{ h \in \mathcal{J} \,\middle|\, \frac{1}{n} \sum_{t=1}^{n} h(x_t) \leq r \right\},$$

and the *local Rademacher complexity* is

$$\mathfrak{R}_n(\mathcal{J}, r) = \sup_{x_{1:n}} \mathfrak{R}_n(\mathcal{J}[r, x_{1:n}], x_{1:n}).$$

For fixed $n \in \mathbb{N}$, we define a *localization function* for $\mathcal{J}$ as any function $\phi_n : [0, \infty) \to \mathbb{R}$ satisfying the following conditions:

1. $\phi_n(r)$ is nonnegative;
2. $\phi_n(r)$ is nondecreasing;
3. $\phi_n(r)$ is strictly positive for some r;
4. $\phi_n(r)/\sqrt{r}$ is nonincreasing;

along with, for all $r \geq 0$,

$$\mathfrak{R}_n(\mathcal{J}, r) \leq \phi_n(r).$$

A size-n *localization radius* of $\mathcal{J}$ with respect to a localization function ϕ_n is the largest real number $r_n^* \geq 0$ satisfying $r_n^* = \phi_n(r_n^*)$.

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4.2. *Proof for risk upper bound* (*Theorem* 1). The main result for this section is the following theorem, which is a generalization of Theorem 1 stated in terms of fixed points for local Rademacher complexity.

THEOREM 3 (Quantitative version of Theorem 1). *For any $\mathcal{F} \subseteq \mathcal{K}(\mathcal{X}, \mathcal{Y})$, $n > 1$, $\varepsilon > 1/\sqrt{n}$, and localization function ϕ_n for $\mathcal{F}^{\mathrm{H}} = \{x \mapsto d_{\mathrm{H}}^2(f(x), g(x)) | f, g \in \mathcal{F}\}$, the corresponding size-$n/2$ localization radius $r_{n/2}^*$ satisfies*

$$\sup_{\mu_{\mathcal{X}}} \sup_{f^* \in \mathcal{F}} \mathcal{R}_n(\hat{f}^{\mathrm{agg}}; f^*, \mu_{\mathcal{X}}) \\ \leq \frac{2\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n)}{n+2} + 2[2 + \log(nBK)]\left[2\varepsilon^2 + 106 r_{n/2}^* + \frac{96\log n + 576 \log\log n + 4}{n}\right],$$

where $\hat{f}^{\mathrm{agg}}$ is the estimator defined in Section 3.3.

In order to derive Theorem 1 from this result, it remains to control r_n^*. This is achieved using the following lemma.

LEMMA 5. *In each of the following three cases, one may choose a localization function ϕ_n for $\mathcal{F}^{\mathrm{H}} = \{x \mapsto d_{\mathrm{H}}^2(f(x), g(x)) | f, g \in \mathcal{F}\}$ such that for all $n \geq 1$, the corresponding size-n localization radius r_n^* satisfies*:

(i) *For any $\mathcal{F} \subseteq \mathcal{K}(\mathcal{X}, \mathcal{Y})$,*

$$r_n^* \leq 972(\log 2n)^3 \left(\inf_{\gamma > 0}\left\{4\gamma + \frac{17}{\sqrt{n}} \int_\gamma^1 \sqrt{\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \rho/2, n)}\, \mathrm{d}\rho\right\}\right)^2.$$

(ii) *If there exist $p, C_{\mathcal{F},n} > 0$ such that $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n) \leq p \log(C_{\mathcal{F},n}/\varepsilon)$, then for*

$$r_n^* \leq \frac{289p}{n} \log\left(\frac{4C_{\mathcal{F},n}\sqrt{n}}{17\sqrt{p}}\right).$$

(iii) *If $|\mathcal{F}| < \infty$,*

$$r_n^* \leq \frac{289 \log |\mathcal{F}|}{n}.$$

We now prove Theorem 3, deferring the proof of Lemma 5 to Appendix A of the Supplementary Material [10].

PROOF OF THEOREM 3. Fix $\mu_{\mathcal{X}}$ and $f^* \in \mathcal{F}$. First, we observe that conditioned on the (independent) sample $(x'_{1:n}, y'_{1:n})$ used to define the empirical Hellinger cover $\mathcal{G}$, Jensen's inequality gives

$$\begin{aligned} &\mathbb{E}_{x_{1:n}\sim\mu_{\mathcal{X}}} \mathbb{E}_{y_{1:n}\sim f^*(x_{1:n})} \mathbb{E}_{x\sim\mu_{\mathcal{X}}} d_{\mathrm{KL}}^2(f^*(x), \hat{f}_n^{\mathrm{agg}}(x)) \\ &\quad = \mathbb{E}_{x_{1:n}\sim\mu_{\mathcal{X}}} \mathbb{E}_{y_{1:n}\sim f^*(x_{1:n})} \mathbb{E}_{x\sim\mu_{\mathcal{X}}} d_{\mathrm{KL}}^2\left(f^*(x), \frac{1}{n+1}\sum_{t=0}^n \hat{f}_t(x)\right) \\ &\quad \leq \mathbb{E}_{x_{1:n}\sim\mu_{\mathcal{X}}} \mathbb{E}_{y_{1:n}\sim f^*(x_{1:n})} \mathbb{E}_{x\sim\mu_{\mathcal{X}}} \frac{1}{n+1}\sum_{t=0}^n d_{\mathrm{KL}}^2(f^*(x), \hat{f}_t(x)) \\ &\quad = \mathbb{E}_{x_{1:n+1}\sim\mu_{\mathcal{X}}} \mathbb{E}_{y_{1:n+1}\sim f^*(x_{1:n+1})} \frac{1}{n+1} \log\left(\frac{\prod_{t=0}^n [f^*(x_{t+1})](y_{t+1})}{\prod_{t=0}^n [\hat{f}_t(x_{t+1})](y_{t+1})}\right). \end{aligned} \tag{3}$$

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Now, observe that the denominator $\prod_{t=0}^{n}[\hat{f}_t(x_{t+1})](y_{t+1})$ is equal to

$$\begin{aligned}
&\Bigl(\sum_{g\in\mathcal{G}}\frac{[\widetilde{g}^{(\alpha)}(x_1)](y_1)}{|\mathcal{G}|}\Bigr)\Bigl(\frac{\sum_{g\in\mathcal{G}}\prod_{t=1}^{2}[\widetilde{g}^{(\alpha)}(x_t)](y_t)}{\sum_{g\in\mathcal{G}}[\widetilde{g}^{(\alpha)}(x_1)](y_1)}\Bigr)\cdots\Bigl(\frac{\sum_{g\in\mathcal{G}}\prod_{t=1}^{n+1}[\widetilde{g}^{(\alpha)}(x_t)](y_t)}{\sum_{g\in\mathcal{G}}\prod_{t=1}^{n}[\widetilde{g}^{(\alpha)}(x_t)](y_t)}\Bigr)\\
&\quad=\frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}}\prod_{t=1}^{n+1}[\widetilde{g}^{(\alpha)}(x_t)](y_t)\ge\frac{1}{|\mathcal{G}|}\prod_{t=0}^{n}[\widetilde{g}^{(\alpha)}_{f^*}(x_{t+1})](y_{t+1}),
\end{aligned}\tag{4}$$

where we define

$$g_{f^*}=\operatorname*{arg\,min}_{g\in\mathcal{G}}\frac{1}{n}\sum_{t=1}^{n}d_{\mathrm{H}}^2\bigl(f^*(x'_t),g(x'_t)\bigr).$$

Thus, since $\widetilde{g}^{(\alpha)}_{f^*}$ only depends on $x'_{1:n}$, $\mathcal{R}_n(\hat{f}^{\mathrm{agg}};f^*,\mu_{\mathcal{X}})$ is bounded above by

$$\begin{aligned}
&\mathbb{E}_{x'_{1:n}\sim\mu_{\mathcal{X}}}\frac{\log|\mathcal{G}|}{n+1}+\mathbb{E}_{x'_{1:n}\sim\mu_{\mathcal{X}}}\mathbb{E}_{x_{1:n+1}\sim\mu_{\mathcal{X}}}\mathbb{E}_{y_{1:n+1}\sim f^*(x_{1:n+1})}\frac{1}{n+1}\sum_{t=1}^{n+1}\log\Bigl(\frac{[f^*(x_t)](y_t)}{[\widetilde{g}^{(\alpha)}_{f^*}(x_t)](y_t)}\Bigr)\\
&\quad\le\frac{\mathcal{H}_{\mathrm{H},2}(\mathcal{F},\varepsilon,n)}{n+1}+\mathbb{E}_{x'_{1:n}\sim\mu_{\mathcal{X}}}\mathbb{E}_{x\sim\mu_{\mathcal{X}}}d_{\mathrm{KL}}^2\bigl(f^*(x),\widetilde{g}^{(\alpha)}_{f^*}(x)\bigr).
\end{aligned}$$

Further, by Lemmas 4 and 3, for all $x\in\mathcal{X}$

$$d_{\mathrm{KL}}^2\bigl(f^*(x),\widetilde{g}^{(\alpha)}_{f^*}(x)\bigr)\le 2\bigl[2+\log(2BK/\alpha)\bigr]\bigl[d_{\mathrm{H}}^2\bigl(f^*(x),g_{f^*}(x)\bigr)+\alpha\bigr].$$

We now make use of a uniform concentration guarantee for the Hellinger distance, which follows by applying Theorem 6.1 of Bousquet [18] (a generic concentration guarantee based on local Rademacher complexity) to $\mathcal{F}^{\mathrm{H}}$ and using that $\sup_{h\in\mathcal{F}^{\mathrm{H}}}\sup_{x\in\mathcal{X}}h(x)\le 1$.

LEMMA 6. *For any localization function ϕ_n for $\mathcal{F}^{\mathrm{H}}$ with corresponding localization radius r_n^*, with probability at least $1-\delta$ over $x'_{1:n}$, all $f,g\in\mathcal{F}$ satisfy*

$$\mathbb{E}_{x\sim\mu_{\mathcal{X}}}d_{\mathrm{H}}^2\bigl(f(x),g(x)\bigr)\le\frac{2}{n}\sum_{t=1}^{n}d_{\mathrm{H}}^2\bigl(f(x'_t),g(x'_t)\bigr)+106r_n^*+48\frac{\log(1/\delta)+6\log\log n}{n}.\tag{5}$$

In particular, since Lemma 6 holds uniformly for all $f,g\in\mathcal{F}$, it can be applied to $f=f^*$ and $g=g_{f^*}$. Converting the resulting guarantee to a moment bound by setting $\delta=1/n$ and using boundedness of $\mathcal{F}^{\mathrm{H}}$, we find

$$\mathbb{E}_{x'_{1:n}\sim\mu_{\mathcal{X}}}\mathbb{E}_{x\sim\mu_{\mathcal{X}}}d_{\mathrm{H}}^2\bigl(f^*(x),g_{f^*}(x)\bigr)\le 2\varepsilon^2+106r_n^*+48\frac{\log n+6\log\log n}{n}+\frac{1}{n}.$$

We obtain the desired theorem statement by taking $\alpha=1/n$ and recalling that each n actually corresponds to "$n/2$", since we split the sample in half. □

4.3. *Proof for risk lower bound* (*Theorem* 2). We now proceed to state and prove Theorem 4, which is a generalization of our main lower bound, Theorem 2. Our general result is most directly stated in terms of *local empirical entropy*. Informally, the local entropy for a given function class captures the complexity of the class when we restrict to an ε-ball (with respect to empirical Hellinger distance) around a given target function.

DEFINITION 1. A set $\overline{\mathcal{G}}^{\mathrm{loc}} \subseteq \mathcal{F}$ is said to be a *local* (d,q)*-packing of* $\mathcal{F}$ *at* $f \in \mathcal{F}$ *on* $x_{1:n} \subseteq \mathcal{X}$ *at scale* ε if

$$\inf_{g' \neq g \in \overline{\mathcal{G}}^{\mathrm{loc}}} \left(\frac{1}{n} \sum_{t=1}^{n} d^q\big(g'(x_t), g(x_t)\big) \right)^{1/q} \geq \varepsilon/2$$

and

$$\sup_{g \in \overline{\mathcal{G}}^{\mathrm{loc}}} \left(\frac{1}{n} \sum_{t=1}^{n} d^q\big(f(x_t), g(x_t)\big) \right)^{1/q} \leq \varepsilon.$$

We denote the cardinality of the largest local packing $\overline{\mathcal{G}}^{\mathrm{loc}}$ by $\overline{\mathcal{N}}^{f}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n})$. The *local empirical* (d,q)*-entropy of* $\mathcal{F}$ *for samples of length n at scale* ε is then defined as

$$\overline{\mathcal{H}}^{\mathrm{loc}}_{d,q}(\mathcal{F}, \varepsilon, n) = \sup_{f \in \mathcal{F}} \sup_{x_{1:n} \subseteq \mathcal{X}} \log \overline{\mathcal{N}}^{f}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n}).$$

Equipped with this definition, we state the main lower bound.

THEOREM 4 (Quantitative version of Theorem 2). *For any* $\mathcal{F}$ *and* $n > 1$,

$$\mathcal{R}_n(\mathcal{F}) \geq \sup_{\varepsilon > n^{-1/2}(\log n)^{-1}} \frac{\varepsilon^2}{4}\left[1 - \frac{12n[1 + \log(2BKn\log n)]\varepsilon^2 + \log(2)}{\overline{\mathcal{H}}^{\mathrm{loc}}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n)}\right].$$

Theorem 2 readily follows by combining this result with the following empirical variant of a result of Yang and Barron [80], Lemma 3, which allows us to pass from local to global entropy.

LEMMA 7. *For all* $q \geq 1$, $\varepsilon > 0$, *and* $x_{1:n} \subseteq \mathcal{X}$,

$$\overline{\mathcal{H}}_{d,q}(\mathcal{F}, \varepsilon/2, x_{1:n}) - \overline{\mathcal{H}}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n}) \leq \overline{\mathcal{H}}^{\mathrm{loc}}_{d,q}(\mathcal{F}, \varepsilon, x_{1:n}) \leq \overline{\mathcal{H}}_{d,q}(\mathcal{F}, \varepsilon/2, x_{1:n}).$$

The remainder of this section is spent proving Theorem 4; we defer the detailed derivation of Theorem 2 to Appendix A of the Supplementary Material [10].

PROOF OF THEOREM 4. Let $\overline{x}_{1:n} \subseteq \mathcal{X}$ witness a maximal local packing for $\mathcal{F}$. That is, there exists $f \in \mathcal{F}$ and $\overline{\mathcal{G}}^{\mathrm{loc}} \subseteq \mathcal{F}$ such that $\overline{\mathcal{G}}^{\mathrm{loc}}$ is a local $(\mathrm{H},2)$-packing of $\mathcal{F}$ at $f \in \mathcal{F}$ on $\overline{x}_{1:n} \subseteq \mathcal{X}$ at scale ε, and $\log|\overline{\mathcal{G}}^{\mathrm{loc}}| = \overline{\mathcal{H}}^{\mathrm{loc}}_{\mathrm{H},2}(\mathcal{F}, \varepsilon, n)$. (If the maximal packing size is not obtained, we can repeat the following argument with a limit sequence.)

Taking $\mu_{\mathcal{X}}$ to be uniform on the witness set, we lower bound the minimax risk using Pinsker's inequality and Markov's inequality. Specifically,

$$\begin{aligned}
\mathcal{R}_n(\mathcal{F}) &= \inf_{\hat{f}} \sup_{\mu_{\mathcal{X}}} \sup_{f^* \in \mathcal{F}} \mathbb{E}_{x_{1:n} \sim \mu_{\mathcal{X}}} \mathbb{E}_{y_{1:n} \sim f^*(x_{1:n})} \mathbb{E}_{x \sim \mu_{\mathcal{X}}} d^2_{\mathrm{KL}}\big(f^*(x), \hat{f}_n(x)\big) \\
&\geq \inf_{\hat{f}} \sup_{f^* \in \overline{\mathcal{G}}^{\mathrm{loc}}} \mathbb{E}_{x_{1:n} \sim \mathrm{Unif}(\overline{x}_{1:n})} \mathbb{E}_{y_{1:n} \sim f^*(x_{1:n})} \mathbb{E}_{x \sim \mathrm{Unif}(\overline{x}_{1:n})} d^2_{\mathrm{H}}\big(f^*(x), \hat{f}_n(x)\big) \\
&\geq \frac{\varepsilon^2}{16} \cdot \inf_{\hat{f}} \sup_{f^* \in \overline{\mathcal{G}}^{\mathrm{loc}}} \mathbb{P}_{x_{1:n} \sim \mathrm{Unif}(\overline{x}_{1:n}), y_{1:n} \sim f^*(x_{1:n})} \left[\frac{1}{n} \sum_{t=1}^{n} d^2_{\mathrm{H}}\big(f^*(\overline{x}_t), \hat{f}_n(\overline{x}_t)\big) \geq \frac{\varepsilon^2}{16}\right].
\end{aligned}$$

Fix an estimator $\hat{f}$ and define

$$g_{\hat{f}_n} = \arg\min_{g \in \overline{\mathcal{G}}^{\mathrm{loc}}} \frac{1}{n} \sum_{t=1}^{n} d_{\mathrm{H}}^2\big(g(\overline{x}_t), \hat{f}_n(\overline{x}_t)\big).$$

We claim that, if the estimator $\hat{f}$ has low risk, then $g_{\hat{f}_n}$ identifies the true conditional density $f^* \in \overline{\mathcal{G}}^{\mathrm{loc}}$. Let $f^* \in \overline{\mathcal{G}}^{\mathrm{loc}}$ be arbitrary, and consider the case where

$$\sqrt{\frac{1}{n} \sum_{t=1}^{n} d_{\mathrm{H}}^2\big(f^*(\overline{x}_t), \hat{f}_n(\overline{x}_t)\big)} < \varepsilon/4$$

and $f^* \neq g_{\hat{f}_n}$. Then, by triangle inequality,

$$\begin{aligned}\sqrt{\frac{1}{n}\sum_{t=1}^{n} d_{\mathrm{H}}^2\big(f^*(\overline{x}_t), g_{\hat{f}_n}(\overline{x}_t)\big)} &\leq \sqrt{\frac{1}{n}\sum_{t=1}^{n} d_{\mathrm{H}}^2\big(f^*(\overline{x}_t), \hat{f}_n(\overline{x}_t)\big)} + \sqrt{\frac{1}{n}\sum_{t=1}^{n} d_{\mathrm{H}}^2\big(\hat{f}_n(\overline{x}_t), g_{\hat{f}_n}(\overline{x}_t)\big)} \\ &\leq 2\sqrt{\frac{1}{n}\sum_{t=1}^{n} d_{\mathrm{H}}^2\big(f^*(\overline{x}_t), \hat{f}_n(\overline{x}_t)\big)} < \varepsilon/2.\end{aligned}$$

This inequality contradicts the assumption that $\overline{\mathcal{G}}^{\mathrm{loc}}$ is a local packing on $\overline{x}_{1:n}$, since that requires $\sqrt{\frac{1}{n}\sum_{t=1}^{n} d_{\mathrm{H}}^2(f^*(\overline{x}_t), g_{\hat{f}_n}(\overline{x}_t))} \geq \varepsilon/2$. Thus,

$$f^* \neq g_{\hat{f}_n} \quad \Longrightarrow \quad \sqrt{\frac{1}{n}\sum_{t=1}^{n} d_{\mathrm{H}}^2\big(f^*(\overline{x}_t), \hat{f}_n(\overline{x}_t)\big)} \geq \varepsilon/4.$$

Consequently, we have

$$\begin{aligned}&\sup_{f^* \in \overline{\mathcal{G}}^{\mathrm{loc}}} \mathbb{P}_{x_{1:n} \sim \mathrm{Unif}(\overline{x}_{1:n}), y_{1:n} \sim f^*(x_{1:n})}\left[\frac{1}{n}\sum_{t=1}^{n} d_{\mathrm{H}}^2\big(f^*(\overline{x}_t), \hat{f}_n(\overline{x}_t)\big) \geq \varepsilon^2/16\right] \\ &\quad \geq \sup_{f^* \in \overline{\mathcal{G}}^{\mathrm{loc}}} \mathbb{P}_{x_{1:n} \sim \mathrm{Unif}(\overline{x}_{1:n}), y_{1:n} \sim f^*(x_{1:n})}\big[f^* \neq g_{\hat{f}_n}\big] \\ &\quad \geq \mathbb{P}_{x_{1:n} \sim \mathrm{Unif}(\overline{x}_{1:n}), f^* \sim \mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}}), y_{1:n} \sim f^*(x_{1:n})}\big[f^* \neq g_{\hat{f}_n}\big].\end{aligned}$$

To proceed, let us recall some standard notation (cf. Section 2 of Cover and Thomas [25]). For any random variables X, Y, the *entropy* of X is $H(X) = -\mathbb{E}_X \log \pi(X)$, the *conditional entropy* is $H(X|Y) = -\mathbb{E}_{X,Y} \log \pi(X|Y)$, and the *mutual information* is $I(X;Y) = H(X) - H(X|Y)$. For another random variable Z, the *conditional mutual information* is $I(X;Y|Z) = H(X|Z) - H(X|Y,Z)$.

Since $x_{1:n}$ and f^* are independent with discrete supports, the joint density $\pi(f^*, x_{1:n}, y_{1:n}) = \pi(y_{1:n}|x_{1:n}, f^*)\pi(x_{1:n})\pi(f^*)$ is well defined. Finally, note that if $f^* \sim \mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})$, $x_{1:n} \sim \mathrm{Unif}(\overline{x}_{1:n})$, and $y_{1:n} \sim f^*(x_{1:n})$, then f^* and $g_{\hat{f}_n}$ are conditionally independent given $(x_{1:n}, y_{1:n})$. Thus, applying Fano's inequality (e.g., Theorem 2.10.1 of [25]), we have

$$\begin{aligned}&\mathbb{P}_{x_{1:n} \sim \mathrm{Unif}(\overline{x}_{1:n}), f^* \sim \mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}}), y_{1:n} \sim f^*(x_{1:n})}\big[f^* \neq g_{\hat{f}_n}\big] \\ &\quad \geq \frac{H(f^*|x_{1:n}, y_{1:n}) - \log(2)}{\log|\overline{\mathcal{G}}^{\mathrm{loc}}|} \qquad (6) \\ &\quad = 1 - \frac{I(f^*; x_{1:n}, y_{1:n}) + \log(2)}{\log|\overline{\mathcal{G}}^{\mathrm{loc}}|}.\end{aligned}$$

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Next, we bound the mutual information $I(f^*; x_{1:n}, y_{1:n})$ using

$$
\begin{aligned}
&\mathbb{E}_{x_{1:n}\sim\mathrm{Unif}(\overline{x}_{1:n}),\, f^*\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}}),\, y_{1:n}\sim f^*(x_{1:n})} \log \frac{\pi(f^*, x_{1:n}, y_{1:n})}{\pi(f^*)\pi(x_{1:n}, y_{1:n})} \\
&\quad = \mathbb{E}_{x_{1:n}\sim\mathrm{Unif}(\overline{x}_{1:n}),\, f^*\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}}),\, y_{1:n}\sim f^*(x_{1:n})} \log \frac{\pi(y_{1:n}|f^*, x_{1:n})}{\mathbb{E}_{g\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})}\pi(y_{1:n}|g, x_{1:n})} \\
&\quad = \mathbb{E}_{x_{1:n}\sim\mathrm{Unif}(\overline{x}_{1:n})}\mathbb{E}_{f^*\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})} d_{\mathrm{KL}}^2\Biggl(\prod_{t=1}^n f^*(x_t), \mathbb{E}_{g\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})}\prod_{t=1}^n g(x_t)\Biggr) \\
&\quad \le \mathbb{E}_{x_{1:n}\sim\mathrm{Unif}(\overline{x}_{1:n})} \inf_{q\in\mathcal{K}(\mathcal{X},\mathcal{Y})} \mathbb{E}_{f^*\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})} d_{\mathrm{KL}}^2\Biggl(\prod_{t=1}^n f^*(x_t), \prod_{t=1}^n q(x_t)\Biggr).
\end{aligned} \tag{7}
$$

The last step holds because for any $q \in \mathcal{K}(\mathcal{X}, \mathcal{Y})$ and $x_{1:n}$,

$$
\begin{aligned}
&\mathbb{E}_{f^*\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})}\Biggl[d_{\mathrm{KL}}^2\Biggl(\prod_{t=1}^n f^*(x_t), \prod_{t=1}^n q(x_t)\Biggr) - d_{\mathrm{KL}}^2\Biggl(\prod_{t=1}^n f^*(x_t), \mathbb{E}_{g\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})}\prod_{t=1}^n g(x_t)\Biggr)\Biggr] \\
&\quad = \mathbb{E}_{f^*\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})}\mathbb{E}_{y_{1:n}\sim f^*(x_{1:n})} \log\Biggl(\frac{\mathbb{E}_{g\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})}\prod_{t=1}^n [g(x_t)](y_t)}{\prod_{t=1}^n [q(x_t)](y_t)}\Biggr) \\
&\quad = d_{\mathrm{KL}}^2\Biggl(\mathbb{E}_{g\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})}\prod_{t=1}^n g(x_t), \prod_{t=1}^n q(x_t)\Biggr) \\
&\quad \ge 0.
\end{aligned}
$$

Then, we bound the last line of equation (7) using chain rule for KL, Jensen's inequality, and inf bounded by sup. That is, for any $\alpha > 0$,

$$
\begin{aligned}
&\mathbb{E}_{x_{1:n}\sim\mathrm{Unif}(\overline{x}_{1:n})} \inf_{q\in\mathcal{K}(\mathcal{X},\mathcal{Y})} \mathbb{E}_{f^*\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})} d_{\mathrm{KL}}^2\Biggl(\prod_{t=1}^n f^*(x_t), \prod_{t=1}^n q(x_t)\Biggr) \\
&\quad \le \inf_{q\in\mathcal{K}(\mathcal{X},\mathcal{Y})} \mathbb{E}_{f^*\sim\mathrm{Unif}(\overline{\mathcal{G}}^{\mathrm{loc}})}\mathbb{E}_{x_{1:n}\sim\mathrm{Unif}(\overline{x}_{1:n})}\sum_{t=1}^n d_{\mathrm{KL}}^2\bigl(f^*(x_t), q(x_t)\bigr) \\
&\quad \le \sup_{g,h\in\overline{\mathcal{G}}^{\mathrm{loc}}} \sum_{t=1}^n \mathbb{E}_{x_t\sim\mathrm{Unif}(\overline{x}_{1:n})} d_{\mathrm{KL}}^2\bigl(g(x_t), \widetilde{h}^{(\alpha)}(x_t)\bigr) \\
&\quad = \sup_{g,h\in\overline{\mathcal{G}}^{\mathrm{loc}}} \sum_{t=1}^n d_{\mathrm{KL}}^2\bigl(g(\overline{x}_t), \widetilde{h}^{(\alpha)}(\overline{x}_t)\bigr).
\end{aligned} \tag{8}
$$

Finally, recall that for any $g, h \in \overline{\mathcal{G}}^{\mathrm{loc}}$,

$$
\sqrt{\sum_{t=1}^n d_{\mathrm{H}}^2\bigl(g(\overline{x}_t), h(\overline{x}_t)\bigr)} \le \sqrt{\sum_{t=1}^n d_{\mathrm{H}}^2\bigl(g(\overline{x}_t), f(\overline{x}_t)\bigr)} + \sqrt{\sum_{t=1}^n d_{\mathrm{H}}^2\bigl(f(\overline{x}_t), h(\overline{x}_t)\bigr)} \le 2\sqrt{n}\varepsilon, \tag{9}
$$

where f is the reference function defining the local packing $\overline{\mathcal{G}}^{\mathrm{loc}}$. Thus, applying equation (9) to equation (8) along with Lemmas 3 and 4 gives

$$
I\bigl(f^*; x_{1:n}, y_{1:n}\bigr) \le 2n\bigl[2 + \log(2BK/\alpha)\bigr]\bigl[2\varepsilon^2 + \alpha\bigr].
$$

Substituting this bound back into equation (6) with $\alpha = 1/[n(\log n)^2]$ gives the result. □

5. On the performance of maximum likelihood. A natural question is whether the estimator introduced in Section 3.3 is required to attain minimax optimal performance, or whether a simpler, more computationally efficient estimator will suffice. In particular, the maximum likelihood estimator (MLE) is simpler both conceptually and computationally. In this section, we highlight the following properties of the MLE for conditional density estimation:

1. For sufficiently small classes—namely, those that satisfy the *Donsker* property (cf. van der Vaart and Wellner [76]) with respect to Hellinger entropy—the MLE obtains the minimax optimal rate. This follows by extending the fundamental results of Shen and Wong [72] and Wong and Shen [78].
2. For richer, nonparametric classes, the MLE can fail to attain the minimax rate. This follows by extending classical results from joint density estimation [15] and nonparametric regression [71]. We also discuss *sieve MLEs* [36], arguing that they are no more efficient than our estimator and their optimality is unknown.

Formally, the maximum likelihood estimator for a sample $(x_{1:n}, y_{1:n})$ is defined as

$$\hat{f}_n^{\mathrm{MLE}} = \operatorname*{arg\,max}_{f\in\mathcal{F}} \sum_{t=1}^{n} \log\bigl([f(x_t)](y_t)\bigr).$$

Our observations in this section also concern the α-smoothed (as defined in Section 3.3) MLE, which we denote by $\widetilde{f}_n^{\mathrm{MLE}(\alpha)}$ for notational convenience.

5.1. *Upper bound on the risk of maximum likelihood.* We first give an upper bound on the performance of the smoothed maximum likelihood estimator, with the proof deferred to Appendix C.1 of the Supplementary Material [10]. To do so, we require a slightly different notion of complexity, the *expected bracketing entropy*, denoted by $\mathcal{H}^{[]}_{d,q}(\mathcal{F},\varepsilon,\mu_{\mathcal{X}})$ (see Appendix C.1 of the Supplementary Material [10]).

It is not hard to see that, in general, $\mathcal{H}_{d,q}(\mathcal{F},\varepsilon,\mu_{\mathcal{X}}) \lesssim \mathcal{H}^{[]}_{d,q}(\mathcal{F},\varepsilon,\mu_{\mathcal{X}})$, and there are cases where the LHS is bounded by a constant yet the RHS tends to infinity. Consequently, our main results in Section 3.2 are generally tighter than the following upper bound. However, in the case when the metric entropy and bracketing entropy agree *and* the model class is sufficiently small, Theorem 5 shows that the MLE is optimal.

THEOREM 5 (MLE upper bound). *For all* $p > 0$,

$$\begin{aligned}
&\mathcal{R}_n\bigl(\widetilde{f}^{\mathrm{MLE}(1/n)}; f^*, \mu_{\mathcal{X}}\bigr)\\
&\quad\lesssim \log(nBK)\cdot \begin{cases} n^{-1/p} & \textit{if } \mathcal{H}^{[]}_{\mathrm{H},2}(\mathcal{F},\varepsilon,\mu_{\mathcal{X}}) \lesssim \varepsilon^{-p} \textit{ for } p>2,\\ n^{-2/(2+p)} & \textit{if } \mathcal{H}^{[]}_{\mathrm{H},2}(\mathcal{F},\varepsilon,\mu_{\mathcal{X}}) \lesssim \varepsilon^{-p} \textit{ for } p\le 2,\\ (p\log n)/n & \textit{if } \mathcal{H}^{[]}_{\mathrm{H},2}(\mathcal{F},\varepsilon,\mu_{\mathcal{X}}) \lesssim p\log(1/\varepsilon). \end{cases}
\end{aligned}$$

Even for *unconditional* density estimation, the tightest results for MLE estimation [78] require conditions on bracketing entropy rather than metric entropy; resolving this gap remains an open problem. Moreover, while it is possible (using Lemma 6) to substitute $\mathcal{H}_{\mathrm{H},2}(\mathcal{F},\varepsilon,n)$ in the above theorem when the bracketing and metric entropies agree, it is unclear how to use a general *empirical* bracketing entropy for all classes. Finally, even in the advantageous setting when empirical Hellinger entropy can be used to characterize the rates, when this scales with ε^{-p} for $p>2$ the upper bound is significantly worse than the optimal rate of $n^{-2/(2+p)}$.

5.2. *The suboptimality of maximum likelihood.* As mentioned above, Theorem 5 provides a suboptimal upper bound for large classes. This suboptimality is not simply an artifact of the analysis, but rather an inherent limitation of the method. For example, Theorem 3 of Birgé and Massart [15] gives an example in which the MLE is suboptimal for joint density estimation, while Example 3 of Shen [71] gives examples in which the MLE is suboptimal for regression with square loss. Recent results have also established the suboptimality of the (closely related) least squares estimator for square loss regression with Gaussian errors and convex function classes [53]. Since conditional density estimation contains both joint density estimation (by setting the covariate space to a singleton) and conditional mean estimation (by setting the errors to be Gaussian) as special cases, these results immediately imply the suboptimality of the MLE for conditional density estimation.

We now present another instance of the suboptimality of the MLE for conditional density estimation. Our example concerns classification with binary responses and covariates, and the purpose is to highlight another natural setting in which the MLE is suboptimal. This construction differs qualitatively from previous results: compared to nonparametric regression the errors are non-Gaussian, and compared to joint density estimation we make nontrivial use of the covariate space. Readers familiar with the construction of Birgé and Massart [15] will recognize the similarities with our analysis, which we defer to Appendix C.2 of the Supplementary Material [10].

THEOREM 6 (MLE lower bound). *For every $\gamma \in (0, 1/2)$, there exists a conditional density class $\overline{\Gamma}_\gamma$ that is convex, contains only γ-Hölder continuous maps from $[-1/2, 1/2]$ to $[7/16, 9/16]$, and has the following property: If the data-generating process is $x_{1:n} \sim \mathrm{Unif}[-1/2, 1/2]^{\otimes n}$ and $y_{1:n} \sim \mathrm{Ber}(1/2)^{\otimes n}$ independently for every $n \in \mathbb{N}$, then for every $\delta > 0$, there exists $C > 0$ such that*

$$\liminf_{n\to\infty} \mathbb{P}_{x_{1:n}, y_{1:n}}\big[\mathbb{E}_{x\sim \mathrm{Unif}[-1/2,1/2]} d_{\mathrm{H}}^2\big(1/2, \overline{G}_{\gamma,n}^{\mathrm{MLE}}(x)\big) \geq C(n \log n)^{-\gamma}\big] \geq 1 - \delta,$$

where $\overline{G}_{\gamma,n}^{\mathrm{MLE}}$ denotes the maximum likelihood estimator for $\overline{\Gamma}_\gamma$.

In particular, we have a well-specified model, yet the risk of the MLE over $\overline{\Gamma}_\gamma$ does not achieve the minimax optimal rate.

Let $\mathcal{F}_\gamma$ denote the set of all γ-Hölder continuous maps from $[-1/2, 1/2]$ to $[7/16, 9/16]$. Then, since $\overline{\Gamma}_\gamma \subseteq \mathcal{F}_\gamma$,[1] Theorem 1 gives

$$\mathcal{R}_n(\overline{\Gamma}_\gamma) \leq \mathcal{R}_n(\mathcal{F}_\gamma) \lesssim n^{-\frac{2\gamma}{2\gamma+1}}. \tag{10}$$

We see that for any $\gamma \in (0, 1/2)$, $n^{-\gamma} > n^{-2\gamma/(2\gamma+1)}$, and hence the maximum likelihood estimator fails to achieve the minimax rate.

Note that in one dimension (of the covariate space), $\gamma = 1/2$ marks the transition point at which $\overline{\Gamma}_\gamma$ ceases to be Donsker with respect to empirical Hellinger entropy. That is, we match the transition point for suboptimality of the MLE that has previously been observed for joint density estimation and square loss regression [15, 71].

We remark that the rate in equation (10) does not match the rate we derive in Example 3 (when $p = 1$, $m = 0$, and $k = 2$), as there is a meaningful difference between the set of Hölder continuous maps to all of $[0, 1]$ and Hölder continuous maps with output bounded away from the boundary. In particular, Hellinger distance behaves linearly rather than quadratically near the boundary of $[0, 1]$, which affects the minimax rate.

[1]For any $p, q \in [7/16, 9/16]$, $(p-q)^2/8 \leq d_{\mathrm{H}}^2(p, q) \leq (p-q)^2$, so that $\mathcal{H}_{\mathrm{H},2}(\mathcal{F}_\gamma, \varepsilon, n) \asymp \mathcal{H}_{2,2}(\mathcal{F}_\gamma, \varepsilon, n) \asymp \varepsilon^{-1/\gamma}$, where the last step follows from, for example, Example 5.11 of Wainwright [77].

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5.3. *Sieve estimators*. A traditional approach to sidestep limitations of the MLE is to use a *sieve* maximum likelihood estimator, which corresponds to the MLE over a smaller subclass of the full model $\mathcal{F}$. Although it is classically known that the sieve MLE can be consistent when the MLE is not (e.g., Geman and Hwang [36]), we now briefly argue that it cannot obtain the minimax rates without sacrificing the computational efficiencies that make the MLE desirable (i.e., the standard MLE does not require construction of a cover). Since the estimate output by the sieve MLE by definition belongs to the subclass used to approximate $\mathcal{F}$, the worst-case risk of the sieve MLE is lower bounded by the scale at which the subclass approximates $\mathcal{F}$ (in expected Hellinger distance). Thus, to achieve the minimax risk, the approximation scale must be chosen at least as small as the minimax risk, so by Theorem 2 the size of the subclass must be at least as large as the cover used by our estimator (described in Section 3.3). Since this requires computing an empirical cover to construct the subclass for the optimal sieve MLE, it is at least as computationally expensive as our estimator.

6. Related work. There is a vast literature on both density estimation and nonparametric regression. The perspective of the present work is to combine (relatively) recent advances in nonparametric regression with classical density estimation techniques to derive minimax rates. In this section, we discuss the most relevant work from both of these lines of research and connect their results to our own. In addition, we discuss connections between our work and contemporary advances in density estimation.

6.1. *Nonparametric density estimation*. The use of metric entropy to study density estimation was pioneered by Le Cam [54], who derived minimax rates for estimation of joint densities with general nonparametric classes using uniform Hellinger entropy; the rates here satisfy the relationship in equation (1). Birgé [11] used a similar approach to obtain minimax rates for bounded metrics on the parameter space, and presented (now classical) examples of the entropy for specific classes of interest. Stone [73] and Ibragimov and Khas'minskii [47] provided minimax rates for smooth nonparametric classes under the L^r loss, and Birgé [12] unified these results and extended them to the Hellinger loss using metric entropy. More recently, entropy conditions for joint density estimation have been used to derive minimax rates for log-concave density estimation [29, 48, 69]. In particular, Kur, Dagan and Rakhlin [52] and Han [44] showed that for this setting, maximum likelihood estimation can achieve optimal rates despite its known suboptimality for certain large classes.

Early lower bounds to complement these results include those of Boyd and Steele [20], who showed that estimation under the L^r loss cannot have a faster rate than $1/n$, and Devroye [27], who showed that for very rich classes (i.e., without regularity and smoothness assumptions) all estimators can be forced to experience an arbitrarily slow rate. Information-theoretic lower bounds based on the fixed point in equation (1) were given by Birgé [11, 12]. Yang and Barron [80] observed that these lower bounds require construction of *local* packings for each class, and instead present a method to derive lower bounds in terms of global entropy. They also provide a simple estimator that obtains the minimax rate for joint density estimation with uniform entropy, which inspired our estimator for conditional density estimation.

While some of the work above instantiates results for conditional density estimation as a special case, there is either the assumption that the conditional distribution follows a simple parametric form, thereby reducing density estimation to conditional mean estimation (e.g., Model 1 of [73]), or the assumption that the covariate distribution is known (e.g., Theorem 6 of [80]). These examples consequently fall under the broader subject of nonparametric regression, which focuses on finding an estimate of the conditional mean $\theta(X) = \mathbb{E}[Y|X]$ rather than the entire conditional density.

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6.2. *Fast rates for nonparametric regression.* The task of estimating the conditional mean (regression function) without specific distributional assumptions is well studied, and comprehensive discussions are given by Györfi et al. [42] and Tsybakov [74]. We wish to highlight a specific line of research into "fast rates" for nonparametric regression that provided significant inspiration for the present work.

Under a sub-Gaussian assumption on the additive regression noise, van de Geer [75] obtained upper bounds for estimation under square loss using empirical entropy conditions. More generally, for bounded, Lipschitz losses (e.g., square loss on a bounded domain), Bartlett and Mendelson [6] provided tighter bounds for the minimax risk of regression using empirical Rademacher complexity [49] rather than the empirical entropy, leading to $n^{-1/2}$ bounds for Donsker classes and $n^{-1/p}$ bounds for larger classes.

Another important advance in nonparametric regression—which enables the tight rates achieved in the present work—is to move from global complexities to localized complexities by adapting modulus-of-continuity techniques (previously used to bound empirical processes [76]) to regression. Koltchinskii and Panchenko [51] and Bousquet, Koltchinskii and Panchenko [19] used this strategy to obtain generalization bounds (and consequently risk bounds for empirical risk minimization) of rate $1/n$ whenever the regression class contains a function that achieves zero empirical error. Bartlett, Bousquet and Mendelson [5] extended these results to noisy regression, where the minimum achievable risk may always be strictly positive. Under a convexity assumption on the regression class, their approach obtains the minimax rate of $n^{-2/(p+2)}$ for regression with Donsker classes under square loss; see Theorem 5.1 of Koltchinskii [50] for more detail. Rakhlin, Sridharan and Tsybakov [65] extended this line of inquiry, using local Rademacher complexities to obtain matching upper and lower bounds for *misspecified* regression under (bounded) square loss for general function classes. Notably, their optimal estimator combines aggregation (like our procedure) with empirical risk minimization (the analogue of maximum likelihood for square loss), and they show that neither procedure can be optimal for all classes on its own in the misspecified setting.

6.3. *Sequential prediction under logarithmic loss.* Conditional density estimation is closely related to the problem of sequential prediction under the logarithmic loss. In this setting, the objective is the same, but the statistician must make predictions one-by-one for individual sequences of data rather than i.i.d. samples. For binary responses and parametric conditional density classes, Rissanen [68], Cover [24], and Barron, Rissanen and Yu [3] obtained nearly exact (including constants) minimax rates. For richer nonparametric classes, Opper and Haussler [61] and Cesa-Bianchi and Lugosi [21] obtained upper bounds on the minimax rates using uniform metric entropies. However, these rates are tight only under strong assumptions on the underlying data generating process (e.g., when densities are bounded uniformly away from zero). As we have discussed for conditional density estimation, the correct notion of complexity in the sequential setting must be *empirical* in nature to avoid penalizing the estimator for the complexity of the entire covariate space simultaneously. For this reason, Rakhlin and Sridharan [63] and Foster et al. [33] use a sequential notion of empirical entropy, introduced by Rakhlin, Sridharan and Tewari [64], to obtain upper bounds on minimax performance. Bilodeau, Foster and Roy [9] refined these arguments to obtain matching upper and lower bounds in terms of the empirical L^∞ entropy. However, their bounds are not tight for *all* possible function classes, which implies that the L^∞ metric is insufficient to completely characterize the minimax rates for logarithmic loss. Bilodeau, Foster and Roy [9] highlighted the setting of Example 1 as an instance for which their rates are not tight; in contrast, our use of the Hellinger distance rather than L^∞ allows us to obtain matching upper and lower bounds for this setting.

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6.4. *Contemporary results on density estimation*. While our results mainly build on the classical nonparametric density estimation literature, density estimation is still an active field of study; we discuss a few of the most relevant recent works here. For conditional density estimation, estimation error has generally only been studied for specific cases. For the L^1 and L^2 losses, this includes kernel-based estimates for certain smoothness classes [17, 31, 41, 43, 46, 57], while for Hellinger risk, this includes exponential family regression [2] and the countable model selection problem for smoothness classes with the assumption that the conditional densities are lower bounded [14, 70].

For KL risk, a natural question is whether our results can easily be extended to the misspecified setting. This appears nontrivial due to the difficulty of controlling likelihood ratios when the measure generating the data does not belong to $\mathcal{F}$, but recent work makes progress toward this goal under additional assumptions. Zhang [82] showed that under a Bernstein condition on the tails of logarithmic loss, the Gibbs posterior (maximum likelihood regularized with the KL divergence) achieves the optimal rates in terms of uniform entropy; for the (bounded) Hellinger risk, no tail conditions are necessary. Cohen and Le Pennec [23] studied penalized maximum likelihood for conditional densities, but only bounded the (nontrivially smaller) Jenson–Kullback–Leibler divergence. Grünwald and Mehta [40] unified a significant line of work [37–39] to provide a generalization of the Bernstein condition that leads to risk bounds for the Gibbs posterior in misspecified density estimation. Finally, Foster et al. [33] exploited properties of *improper estimators* for conditional density estimation with the logistic loss. Mourtada and Gaïffas [60] extended this work to remove $\log(n)$ factors for misspecified conditional density estimation in the parametric setting and obtain improved rates for misspecified Gaussian linear regression.

All existing results for the logarithmic loss or KL risk (and some recent results for Hellinger risk) require tail conditions that hold uniformly over the density class, which immediately rules out a number of basic settings (e.g., univariate, compact response spaces in which densities are arbitrarily close to zero). Such tail conditions are not required to obtain nontrivial worst-case rates: as Cesa-Bianchi and Lugosi [21] noted and exploited, one may always artificially truncate the class of densities to make the logarithmic loss Lipschitz and bounded, then pay an additional penalty for the approximation error incurred by truncation. However, truncation is known to lead to suboptimal rates [63], so it remains an open problem to characterize the optimal rates for misspecified (joint or conditional) density estimation without extra tail conditions.

7. Conclusion. We have characterized the minimax rates for conditional density estimation in terms of empirical Hellinger entropy for the conditional density class. Our results show that the classical fixed point relationship in equation (1) governs the minimax rates for conditional density estimation, provided that one adopts empirical Hellinger entropy as the notion of complexity. Importantly, the empirical Hellinger entropy captures the correct dependence on the complexity of the covariate space, and leads to optimal rates, even for high-dimensional, potentially unbounded, covariate spaces. We leverage new techniques in both density estimation and nonparametric regression to make these new advances.

A natural direction for future work is to extend our results beyond the well-specified setting. While recent work [40, 60] has made progress on providing (excess) risk bounds in the presence of misspecification, it remains an open problem to characterize the minimax rates for conditional density estimation with misspecified models without tail assumptions on the densities.

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SUPPLEMENTARY MATERIAL

Supplement to "Minimax rates for conditional density estimation via empirical entropy" (DOI: 10.1214/23-AOS2270SUPP; .pdf). Additional proofs and extensions to sequential and adaptive estimators.

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